



CV2 Vacuum On-Load Tap-changer

Operating Instructions

HM 0.460.4102-2023



IMPORTANT:

*Please read this manual carefully
before using the tap-changer and
save it for reference.*

Thank you for choosing Huaming products.

Before using, please read this instruction carefully. The product may have been modified since this document was published. We reserve the right to change the technical data, design and scope of supply without prior notice. Generally, information provided with individual quotations and orders is binding. Huaming has no liability for any damage due to incorrect interpretation of this document.

The original document is written in Chinese.

Keep this manual at hand for future use and contact Huaming if you have any questions.

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1. Foreword

Manufacturer

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Subject to change without notice

The information contained in this technical file consists of the technical specifications published at the time of printing, important modifications will be included in the new version of the technical file.

1.1 Safety

Safety instructions are explained in this manual. They must be observed and followed to avoid relevant dangers. Personal safety equipment must be worn during work to prevent potential risks.

	Protective clothing for protecting personnel from hazards in the work environment.
	Safety shoes for protecting personnel from falling sharp objects and slippery surfaces.
	Goggles for protecting eyes and facial area from impact, dust, liquid splash and high heat hazards.
	Hard helmet for protecting head from injury.
	Hearing protection for protecting hearing from damage.
	Protective gloves for protecting hands from mechanical, chemical, and electrical hazards.

Table 1.1.1 Personal safety equipment list

1.2 Security

The following format is used for warnings in this technical paper:



Warning slogan

Adverse consequences of safety violations

Safe behaviours to avoid adverse consequences

The following signal words are used:

Warning slogan	Hazard level	Consequences of Violations
Danger	Immediate threat of danger	Death or serious injury
Warning	Probable threat of danger	Fatal or serious injury
Prohibition	Probable dangerous situation	Minor or moderate injury
Note	Possible dangerous situation	Material damage

Table 1.2.1 Signal words in safety instructions

Warning slogan	Meaning
	The yellow triangle warning sign means life and health may be at risk if neglecting the instruction. Neglecting this warning can lead to serious or fatal injuries.
	The yellow triangle warning sign means equipment and personnel may be at risk of electricity shock if neglecting the instruction. Neglecting this warning can lead to serious or fatal injuries.
	The yellow warning triangle sign means that if the indication is ignored, equipment and personnel can be exposed to fire, deflagration, etc.
	General prohibition signs! The red symbol indicates that the behaviour is prohibited.

Table 1.2.2 Symbols used in warning slogan

1.2.1 Information system

Information is designed to simplify and improve understanding of particular procedures. In this technical file they are laid out as follows:



NOTE!

The blue note symbol indicates important information that must be noted.

1.2.2 Safety precautions

Precaution	Action
Electric shock	Always unplug a device before inspection. Stay away from high-voltage equipment and wear protective clothing.
Transformer oil	Collect used transformer oil in dedicated drums. Keep it away from flames or sparks. Do not touch it which may cause damages to health.
Fire	In case of fire, extinguish it with powder, foam, or carbon dioxide. Do not use water.
Waste	Wastes are dangerous to the environment or human health. Clean them up in time and use different methods according to the wastes' categories.

Table 1.2.3 Safety precaution

1.2.3 Appropriate operation

The product and associated equipment and special tools supplied with it comply with the relevant legislation, regulations, and standards, particularly health and safety requirements, applicable at the time of delivery.

If used as intended in compliance with the specified requirements and conditions in this technical file as well as the warning notices in this technical file and attached to the product, then the product does not present any hazards for persons, property, or the environment. This applies during the entire lifespan, from delivery through installation and operation to disassembly and disposal.

The operational quality assurance system ensures a consistently high-quality standard, particularly when it comes to observance of the health and safety requirements.

The following operations are considered as compliant operations:

- 1) The products are to be operated in accordance with this technical file, the agreed delivery conditions and the technical data, and the associated equipment and special tools supplied with the products are to be used exclusively for the intended purpose and in accordance with the specifications of this technical file.
- 2) This product can only be used with the transformer specified in the order. The serial number of the motor-drive unit corresponds to that of the CV2.

1.2.4 Inappropriate use

Operating not performed in accordance with Chapter 1.2.3 will be considered inappropriate.

Shanghai Huaming Power Equipment Co., Ltd does not accept liability for damage from unauthorized or inappropriate changes to the product. Unauthorized changes to the product without consultation with Huaming can lead to personal injury, material damage and operational faults.

1.2.5 Personnel qualification

The product is designed only for use in electrical or energy systems and facilities operated by appropriately trained staff. This staff comprises people who are familiar with the installation, assembly, commissioning, and operation of such products.

1.2.6 Obligation of the operator

To prevent accidents, faults and damage as well as unacceptable adverse effects on the environment, those responsible for transport, installation, operation, maintenance and disposal of the product or parts of the product must ensure the following:

- 1) All warning and hazard notices are complied with.
- 2) Personnel are instructed regularly in all relevant aspects of operational safety, the Operating Instructions and particularly the safety instructions contained therein.
- 3) Regulations and operating instructions for safe working as well as the relevant instructions for staff procedures in the case of accidents and fires are kept on hand at all times and displayed in the workplace where applicable.
- 4) The product is only used in a sound operational condition and safety equipment in particular is checked regularly for operational reliability.
- 5) Use only replacement parts, lubricants and aids which are authorized by the manufacturer.
- 6) Comply with the specified operating conditions and requirements of the installation location.
- 7) All necessary devices and personal protective equipment for each activity are available.
- 8) The prescribed maintenance intervals and the relevant regulations are complied with.
- 9) Installation, electrical connection and commissioning of the product is only carried out by qualified and trained personnel in accordance with this technical file.
- 10) The operator must ensure appropriate use of the product.

1.3 General

The CV2 on-load tap-changer is mounted on the transformer cover using the head flange and is connected to the motor-drive unit via its top gear, reversing gear box (accessory) and drive shaft (horizontal and vertical) to achieve motor-driven or remote-controlled tap-change operations (see Figure 1.3.1).

The on-load tap-changer type CV2 is a selector switch. (see Figure 1.3.2).

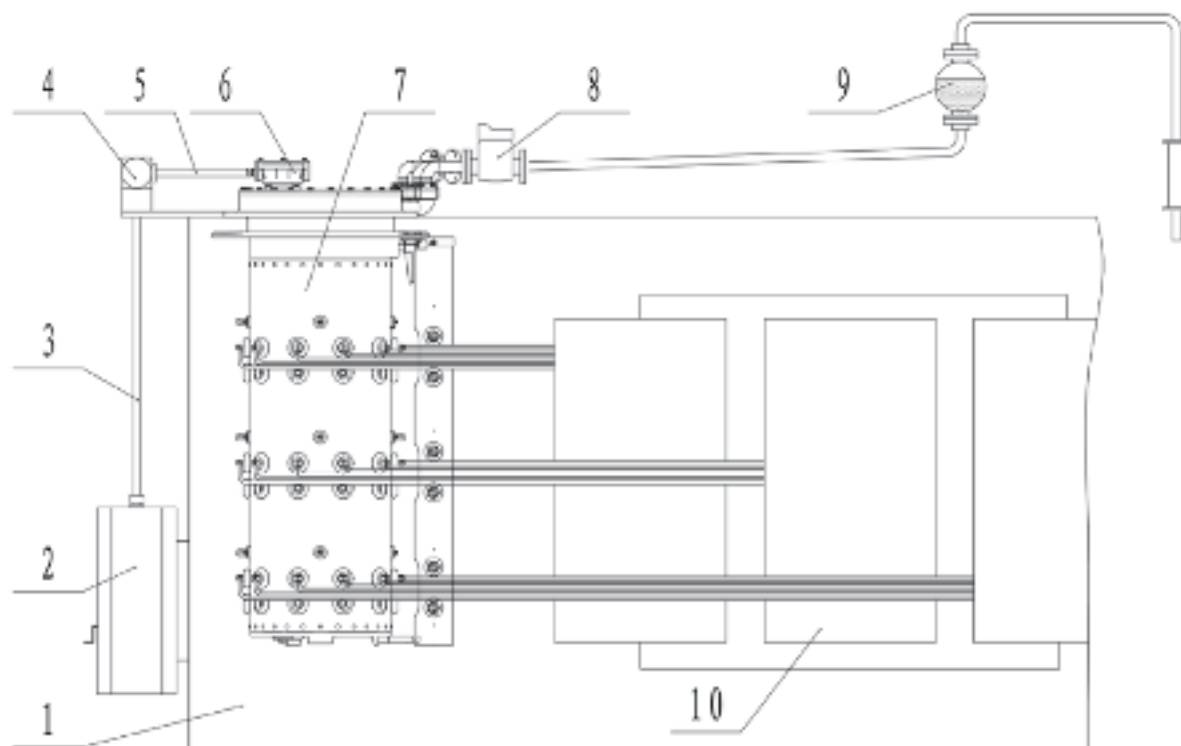


Figure 1.3.1 Overview of tap-changer inside transformer

1. Transformer tank	2. Transformer tank
3. Vertical drive shaft	4. Vertical drive shaft
5. Horizontal drive shaft	6. Horizontal drive shaft
7. Tap-changer	8. Tap-changer
9. Oil conservator	10. Oil conservator

Table 1.3.1 List of all parts

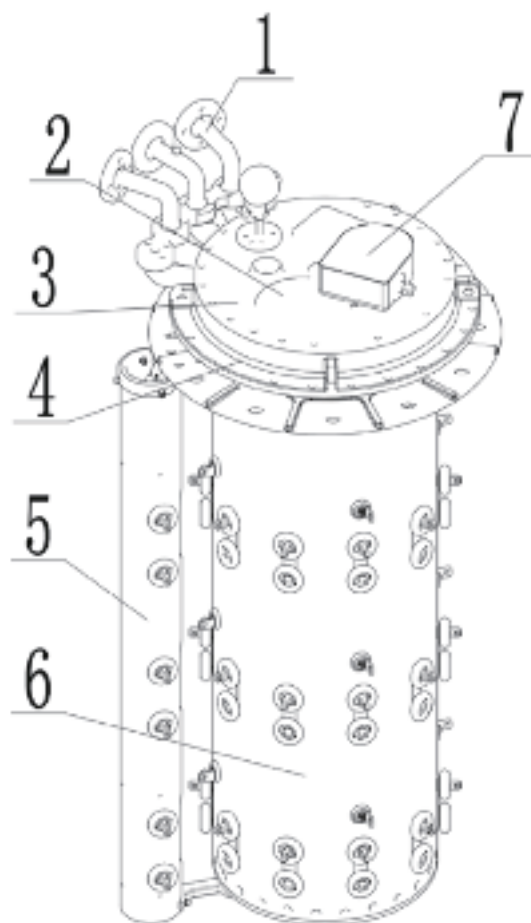


Figure 1.3.2 Overview of tap-changer

1. Oil pipe	2. Rupture disk
3. Tap-changer head cover	4. Tap-changer head
5. Tap selector	6. Oil compartment
7. Top gear	

Table 1.3.2 List of tap-changer components

2. Technical data

2.1 Technical data

Table 2.1.1 Technical Data

item	Description		CV2III		CV2I	
1	Maximum Rated though Current (A)		350	600	350	600
2	Rated frequency (Hz)		50 or 60			
3	connection		Any connection			
4	Maximum rated class voltage (V)		1500-200			
5	Rated class capacity (kVA)		700	750	700	750
6	Short-circuit capacity (kA)	Thermal (3s)	5	8	5	8
		Dynamic (peak)	12.5	20	12.5	20
7	Maximum number of working tap positions		Max 12 Without switch selector; max 23 with switch selector			
8	Insulation level to earth (kV)	The highest voltage for equipment (kV)	40.5	72.5	126	145
		Rated separate source AC withstand voltage (kV/50Hz,1min)	95	140	230	275
		Rated lightning impulse withstand voltage (kV,1.2/50μs)	250	350	550	650
9	mechanical life		1.5 million times			
10	Maintenance-free times		300,000 times			
11	selector switch oil compartment	Service pressure	0.03MPa			
		Leakage test	0.1MPa 24 hours without leakage			
		Over pressure protection	Rupture disc bursts at 300 ± 20% KPa			
		Protective relay	Set oil flow speed at 1.0m/s ± 10%			
12	Motor drive unit		SHM-D or CMA7			

2.2 Operation conditions

- 1) The operating temperature of the tap-changer in oil is $\geq -25^{\circ}\text{C}$ and $\leq 105^{\circ}\text{C}$.
- 2) The storage environment temperature of the tap-changer is $\geq -25^{\circ}\text{C}$ and $\leq 40^{\circ}\text{C}$, and the humidity should be not higher than 85%. If beyond this standard, please mark your requirements before ordering. The tap-changer material and accessories will be specially processed to meet the special requirements of the environment.
- 3) After being mounted to the transformer, the non-perpendicular degree between the tap-changer and the ground should be less than 2%.
- 4) No sever dust and other explosive or corrosive gases on the installation site.



Warning!

Condensed water droplets can cause degraded electrical performance, damage to equipment, and can also lead to explosion.

Do not store the tap-changer or motor drive unit in a place with high humidity ($\geq 85\%$). This will cause condensation.



NOTE!

Special design is available to meet particular environment conditions. Please contact Huaming for further information.

2.3 Type designation

2.3.1 Type representation

CV2 tap-changers have a variety of specifications due to different combinations of the number of phases, the maximum rated through current, the highest voltage of the equipment, the insulation class of the selector and the connection mode. Therefore, the above-mentioned performance parameters should be clearly reflected in the model of the tap-changer, and the detailed description of each code is shown in Figure 2.3.1.

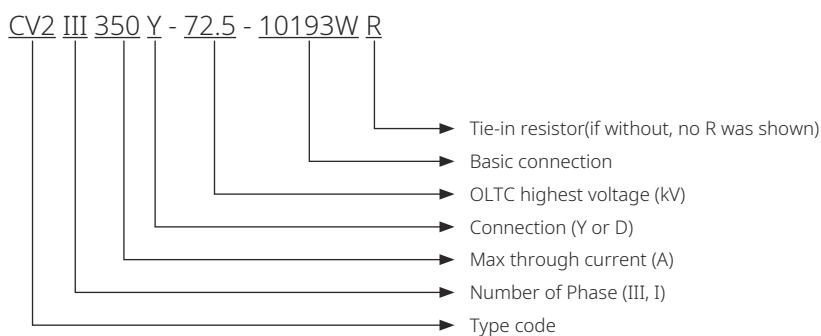


Figure 2.3.1 Type representation

2.3.2 Basic connection

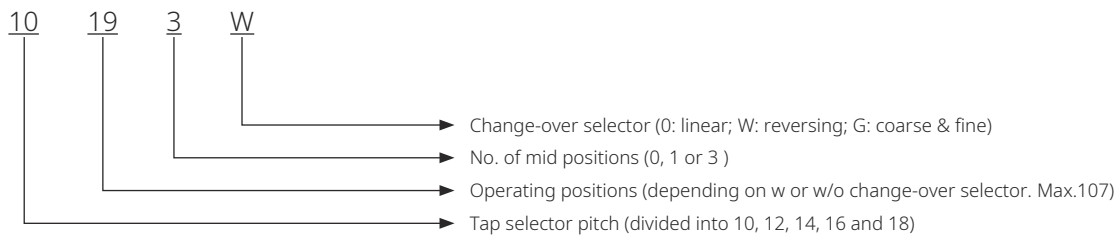


Figure 2.3.2 Tap selector basic cone

- 1) For example, 10090 means that the number of tap selector distribution contacts is 10 and the maximum number of operating positions is 9, and 0 means a linear connection OLTC without change-over selector.
- 2) For example, 10193W indicates that the tap selector has a distribution contact number of 10, a maximum number of operating positions of 19, and an intermediate position of 3. W indicates a selector with polarity changeover.
- 3) For example, 10191G indicates that the tap selector has a distribution contact number of 10, a maximum number of operating positions of 19, and an intermediate position of 1. G indicates a selector with coarse changeover.

3. Packing, Transportation & Storage

3.1 Supplied components

The complete set of tap-changer contains below parts

- 1) Selector switch
- 2) Motor drive unit (MDU), controller and connecting cables,;
- 3) Drive shafts, couplings, bevel gear boxes and other transmission accessories
- 4) Protection relay and accessories
- 5) Special tools and accessories for tap-changer assembly
- 6) Customer-required special accessories if any.

Please refer to the corresponding packing list delivered with the tap-changer for exact supplied components.

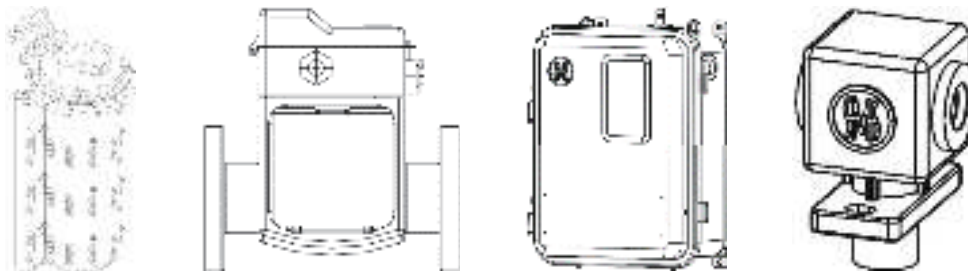


Figure 3.1.1 set of CV2 tap changer

3.2 Transportation & acceptance

3.2.1 Transportation

3.2.1.1 Packings is suitable for all common kinds of transportations. The boxes of tap selector and MDU are allowed to be appropriately stacked during storage, and the load-bearing capacity of the box's top surface shall not exceed 500kg/m².

3.2.1.2 Transportation shall be reasonably arranged according to the center of gravity position marking of the box, and lifting shall be regulated according to the lifting marking on the box.

3.2.1.3 Description of carton warning labelling (Figure 3.2.1)



Figure 3.2.1 Warning signs on the shipping marks

**DANGER!**

The selection of the rope and load reinforcement must be carried out by trained and designated personnel

Do not stand under heavy objects that are hanging.

Use transport tools and lifting devices with a bearing capacity of > 500kg.

3.2.2 Transferring

When the products arrive at the destination, the appropriate method of transfer is selected in accordance with the loading and unloading requirements:

If the transformer is packed in wooden cases, it should be lifted according to the lifting label on the wooden cases. Or transfer according to the label, as shown below.

**NOTE!**

Choose appropriate forklift and crane according to the dimension and gross weight of the boxes.

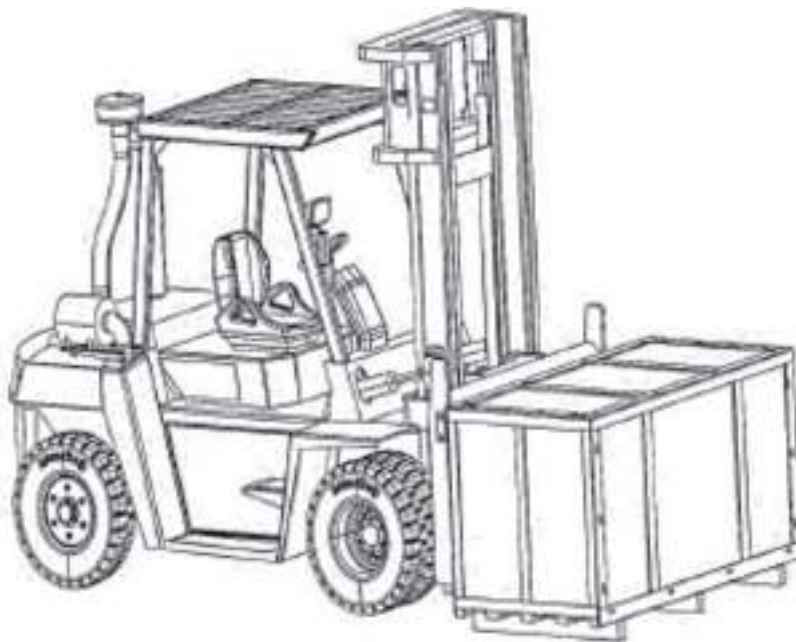


Figure 3.2.2 Wooden case fork - lifting



Figure 3.2.3 wooden case Lifting

3.2.3 Lifting

The OLTC body should be hoisted vertically, and all lifting lugs on the top cover should be fixed before lifting. As shown below:

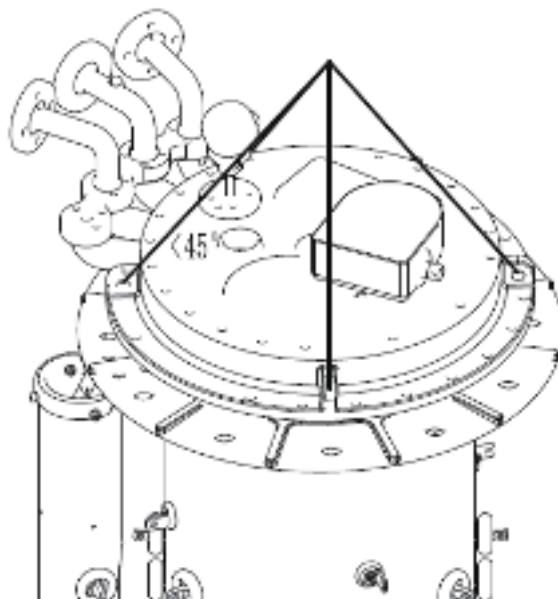


Figure 3.2.4 OLTC Lifting

DANGER!

Damage to equipment or injury to personnel may happen.

At least two horizontal points must be secured during lifting or all lifting points must be fully secured before lifting.

Rapid ascending or descending is prohibited when lifting.

Lifting shall not exceed 45° to prevent overturning.

It is strictly prohibited to stand under the hoisting.



3.2.4 Goods receiving

Receiving Notes:

Each batch of delivered goods must be checked by the consignee before accepted (signing acceptance confirmation). Following items must be checked by the consignee carefully:

- 1) Check the quantity of components received according to the packing list.
- 2) Check the technical data if it is in line with the order.
- 3) Check whether the packing is intact. If it's found to be damaged (if possible, take pictures of the box and damaged goods) or the information of the goods does not match the packing list, consignee should refuse to sign for acceptance, and immediately contact with the shipping company and Huaming.



NOTE!

When receiving the goods, please check carefully the packing and the supplied components according to the packing list. In case anything is broken or missing, please contact Huaming in time.



NOTE!

The transportation and lifting of the box must be operated by professionals.
The carrying capacity of transportation vehicles and cranes should be more than 1000kg.



CAUTION!

The packing box should be firmly fixed during transportation and lifting, and be handled with care to avoid possible damage to the product due to vibration, impact, falling, tipping, bumping, shaking, etc.



CAUTION!

If the box drops or is severely impacted, the possibility of damages must be considered. Comprehensive inspection of the product should be carried out, and if necessary, inspection(repair) shall be done by the Huaming's professional and technical personnel.

The above procedure also applies to corrosion of packaged goods due to moisture (rain, snow, condensation) ingress. OLTC must be stored in a dry place and sealed package before installation.

3.2.5 Acceptance

NOTE!



Avoid damaging the original packaging when unpacking and inspecting.
If the received goods and its accessories do not match the list, contact Huaming in time.
If the box drops or is severely impacted, the possibility of damages must be considered.
Comprehensive inspection of the product should be carried out, and if necessary, inspection(repair) shall be done by the Huaming's professional and technical personnel.

3.3 Storage

When the goods are stored continuously for more than 1 year, the equipment should be inspected thoroughly before installation.

Specially packed goods with sealed packing can be stored outdoors if they meet the following conditions.

- 1) The storage environment temperature should not be $\geq -25^{\circ}\text{C}$ and $\leq 40^{\circ}\text{C}$. If out of this range, special material is required, please specify when ordering.
- 2) The tap-changer must not be stored in an environment with flammable, explosive and corrosive gases.
- 3) It must be protected from moisture, dust, rodents and ants.
- 4) Regularly check whether the stored equipment is abnormal.
- 5) If the tap-changer will be stored for a long time, the desiccant must be replaced regularly, and re-packed with sealed packing.



NOTE!

After long-term storage, the tap-changer and its accessories must be carefully inspected before installation.

4. Installation of the On-load Tap-changer on the Transformer



DANGER!

Improper operation can cause damage to the tap-changer.

Please be sure the shielding caps to be installed on the bolts of the equipment above 35KV during installation.

4.1 Flanges for fixing tap-changer on the transformer case

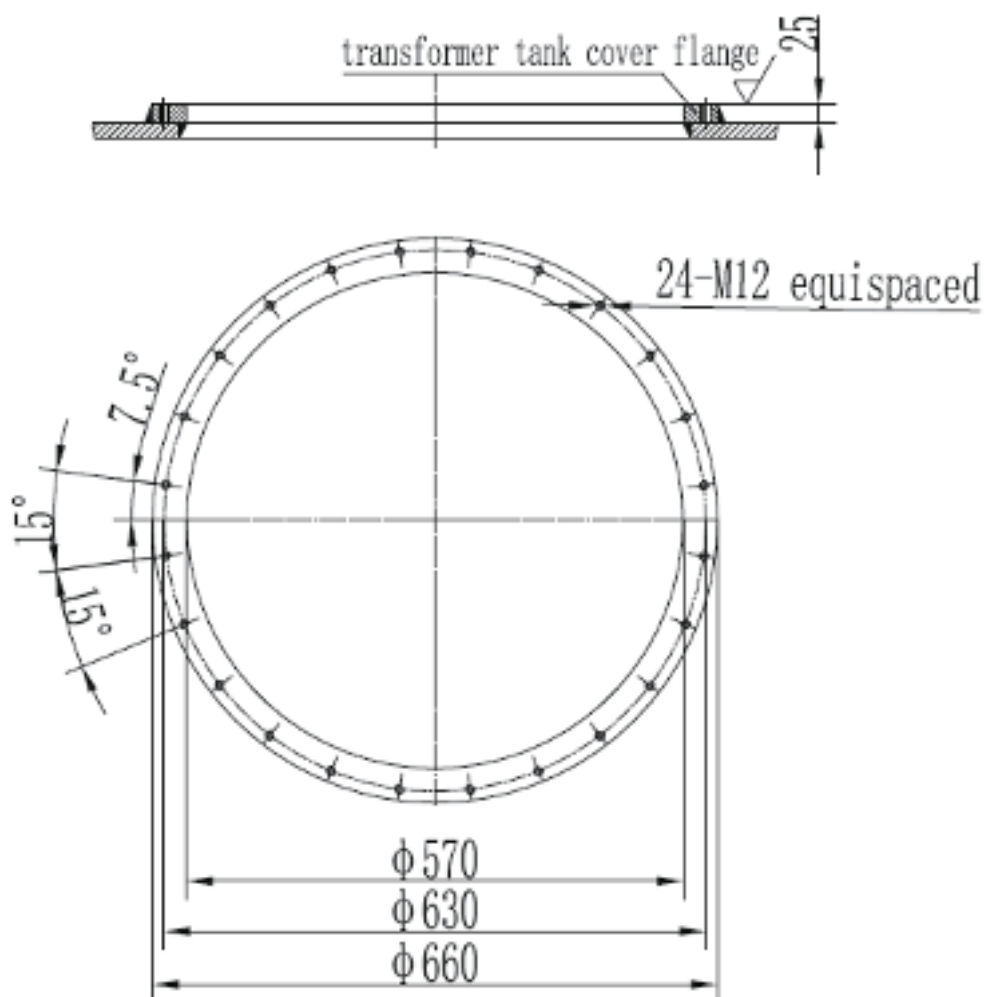


Figure 4.1.1 Flanges for fixing OLTC

Mounting flanges are required for fixing the tap-changer head to the transformer cover. The flange should be constructed in accordance with the shape of the sealing surface of the tap-changer head (see Figure-4.1.1).

4.2 Installation of the on-load tap-changer on a bell-type transformer

4.2.1 tap-changer head cover removal.

4.2.1.1 Make sure that the tap-changer is in the setting position.

4.2.1.2 Unscrew the 24 pieces of M10x35 bolts (item 2) During the removal of tap-changer head cover, avoid damaging the sealing ring of the head cover (Figure-4.2.2).

4.2.1.3 Avoid damaging the sealing surfaces of the tap-changer head cover and head (O-ring, Figure-4.2.2) during disassembly and other operations.

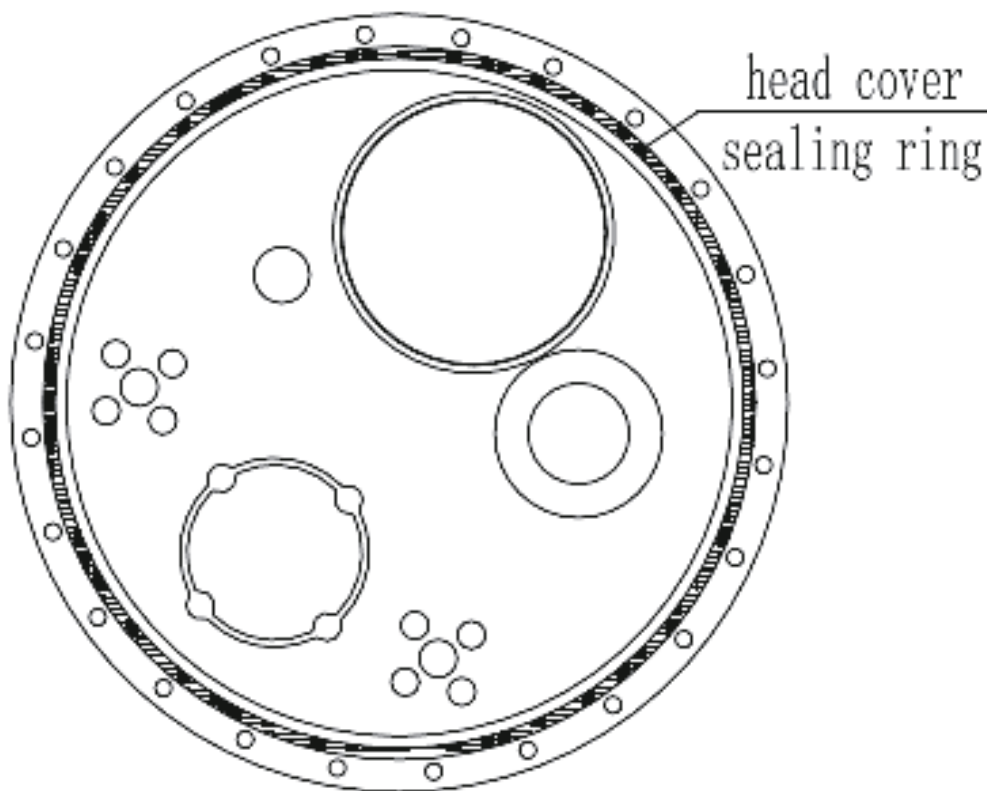


Figure 4.2.1 Schematic diagram of setting position



DANGER!

Damage to equipment and injury to personnel may result.

If the part is not placed on a flat surface, it may tilt and cause injury!

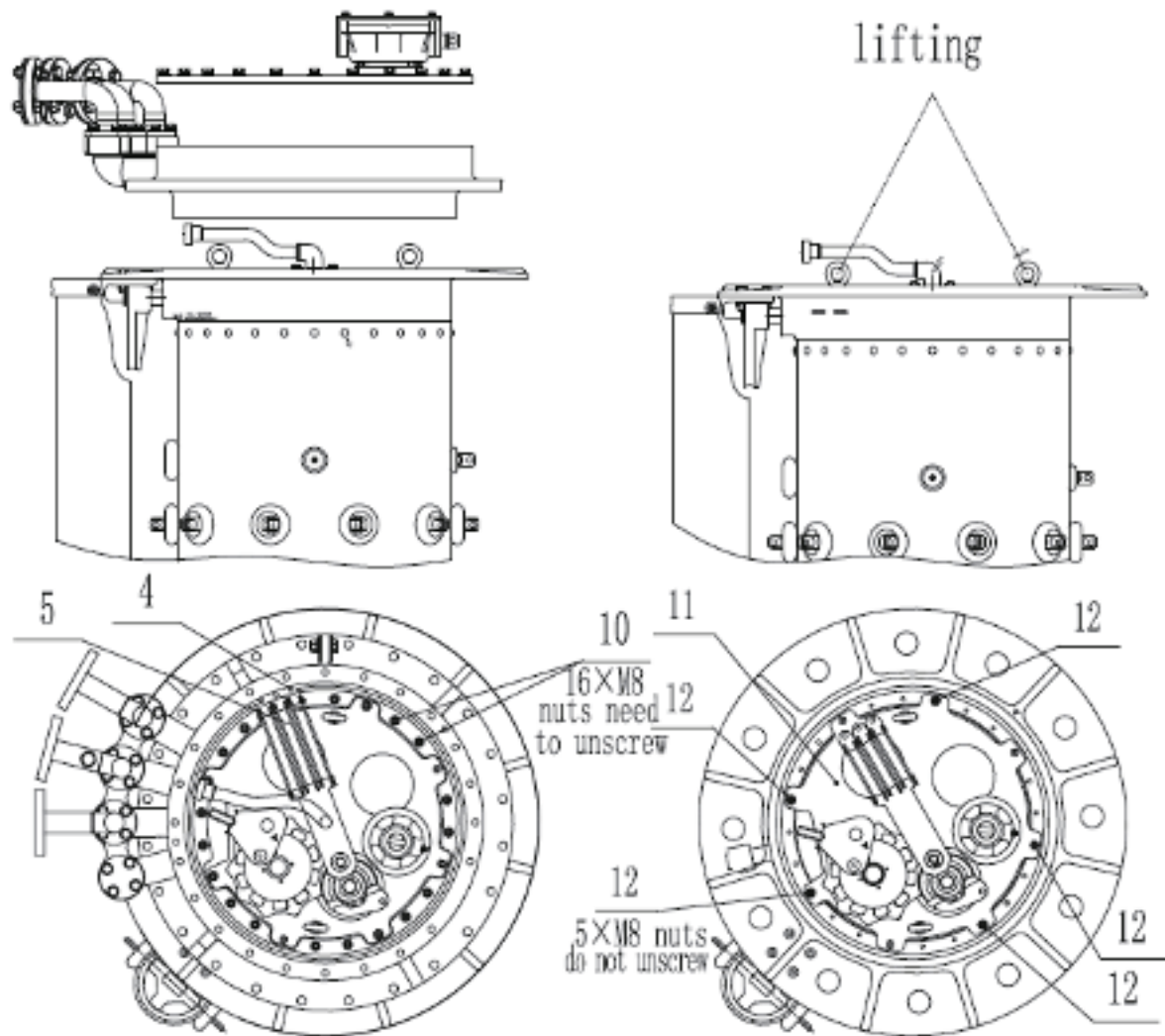
Therefore they must be placed on a horizontal flat surface.

Avoid damaging the sealing surfaces of the tap-changer head cover and head (O-ring, Figure-4.2.2) during disassembly and other operations.



NOTE!

Keep the head cover seal well preserved.



1. head cover 2. bolt M10 3. upper flange 4. elastic pin 5. energy storage spring 6. pipe nut
 7. oil return pipe 8. supporting flange 9. OLTC 10. nut M8 11. the mount 12. nut M8
 13 lifting ring nut

Figure 4.2.2

4.2.2 Remove the energy accumulation system spring (Figure-4.2.2)

Remove 3 pieces of elastic pins (No. 4), rotate energy accumulation spring (No. 5) and make it separate with upper flange (No. 3).

4.2.3 Remove the oil suction pipe (Figure-4.2.2)

Loosen the pipe nuts (No. 6) with adjustable spanner, rotate oil suction pipe (No. 7) to deflect it away from its original position to facilitate the removal of the upper flange.

4.2.4 Remove the upper flange (Figure-4.2.2)

Unscrew 16 pieces of M8 nuts (No. 10) on upper flange (No. 3) and remove the upper flange. Keep all removed parts well preserved for reinstallation.

4.2.5 Installation of the on-load tap-changer on standard type transformer (Figure-4.2.3)

First place the on-load tap-changer on the temporary support frame inside the transformer.

The on-load tap-changer must be in the accurate vertical position on the supporting structure (max. 2° deviation from vertical position). The on-load tap-changer must be mounted on the supporting structure in such a way that the on-load tap-changer can be lifted by 5 to 20 mm to its final position after the transformer bell-type transformer tank cover is closed.

The assembly steps are as follows:

1. Lift the on-load tap-changer into the support frame and connect it to the voltage regulating winding. The on-load tap-changer shall be placed stably on the support frame and must not be moved around.
2. Temporarily fix the on-load tap-changer to the support frame using the mounting holes on the supporting flange. If necessary, put temporary adjusting pad between the temporary support frame and the supporting flange and adjust the mounting height of the on-load tap-changer to the final mounting position. The adjusting pad must be removed before installation of the transformer bell-type tank cover.

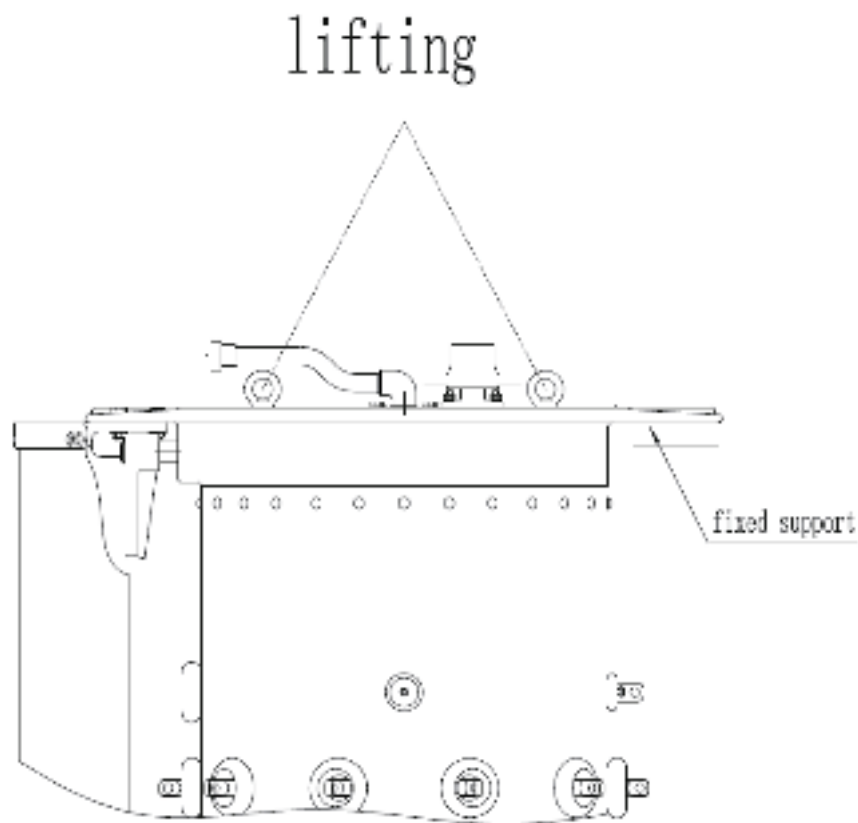


Figure 4.2.3

4.2.6 Assembly of bell-type transformer tank cover

1. Before assembling the bell-type transformer tank cover, clean the sealing surfaces of the supporting flange of the on-load tap-changer oil compartment. Place the sealing ring on the supporting flange (Figure-4.2.4). Pull out the supporting pad.

2. Lift the bell-type transformer tank cover over the transformer body and assemble it.
3. Clean the sealing surface before mounting the on-load tap-changer mounting flange. Place the sealing gasket on the transformer mounting flange and fix the on-load tap-changer mounting flange to the transformer mounting flange (24 pieces of M12 bolts, torque 100 to 110 N·m).

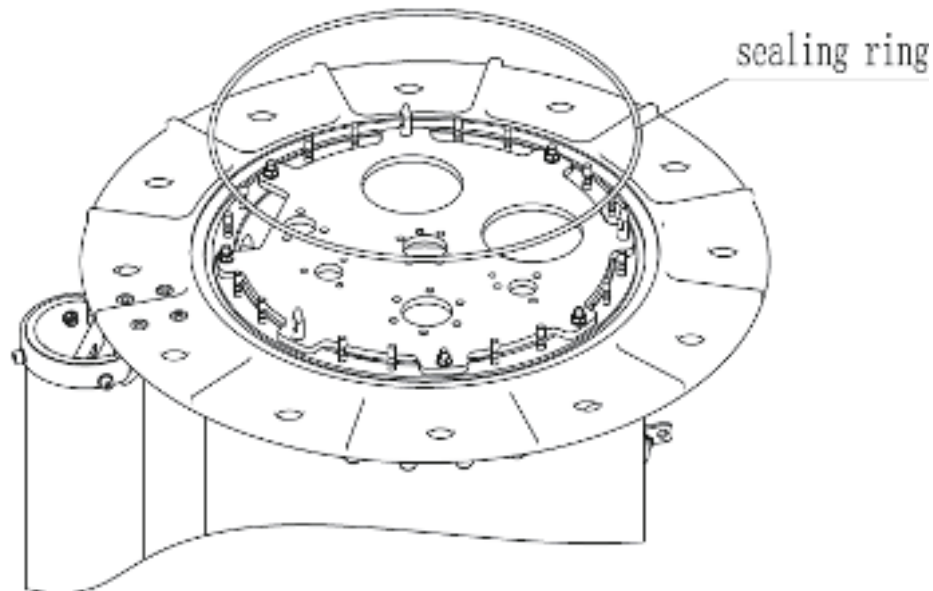


Figure 4.2.4

4. Mount the on-load tap-changer onto the head mounting flange. Use lifting equipment to gently lift the on-load tap-changer body (and insert) upwards from the 2 lifting bolts so that the oil compartment bolts (16 sets of M8 nuts and washers) can be screwed in. Tighten these nuts evenly. (Figure 4.2.5)

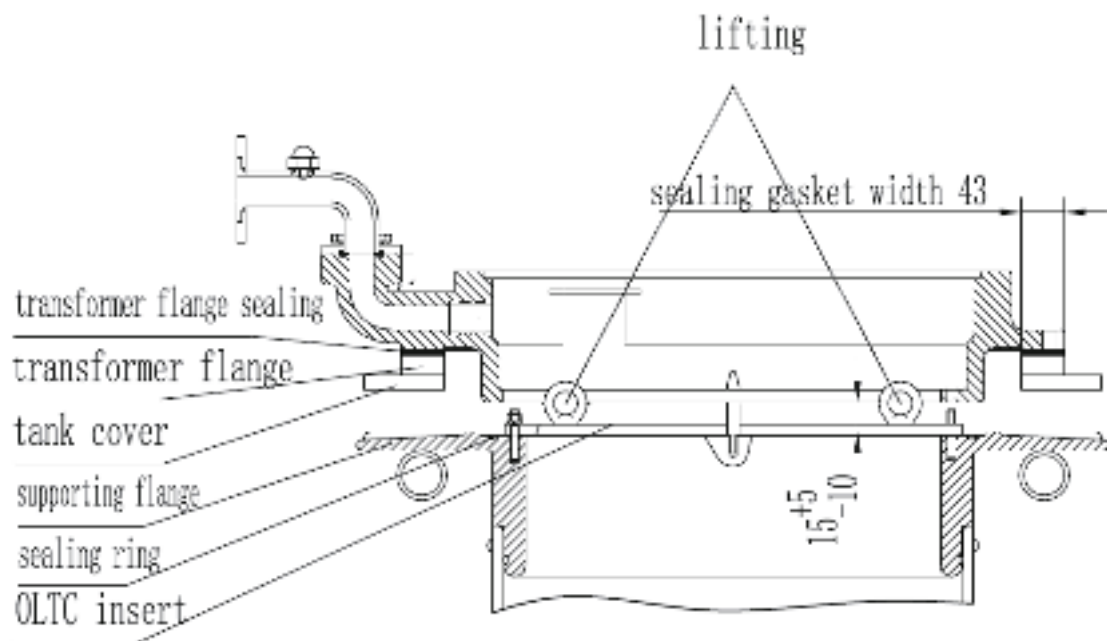


Figure 4.2.5

4.2.7 Reinstallation of the on-load tap-changer energy accumulation spring and oil suction pipe

Proceed the reinstallation of the energy accumulation spring and oil suction pipe in reverse order (4.2.3, 4.2.2) of disassembly.

4.2.8 Installation of top cover

Check the seals before covering the tap changer head cover. Tighten 24 pieces of M10x35 bolts evenly (with washers).



NOTE!

- Do not damage the O-ring seal on the pipe fitting.
- Carefully clean the sealing surface before installing the top cover.

4.3 Installation of the on-load tap-changer on standard type transformer

Please refer to chapter 4.2.1

4.4 Connection between voltage regulating winding tap and on-load tap-changer terminals

The tap selector terminals are provided with through-holes for M10 bolts, which facilitate the connection of transformer voltage regulating winding tap with the tap changer terminals.

Shielding caps for M10 bolts will be supplied (Figure 4.4.1)

4.4.1 Removing the mounting flange

Remove the remaining fixing nuts on the mounting flange (13 pieces of M8 nuts, 13mm spanner, with butterfly washers) and lift the tap-changer mounting flange from the support flange (see Figure 4.2.4).

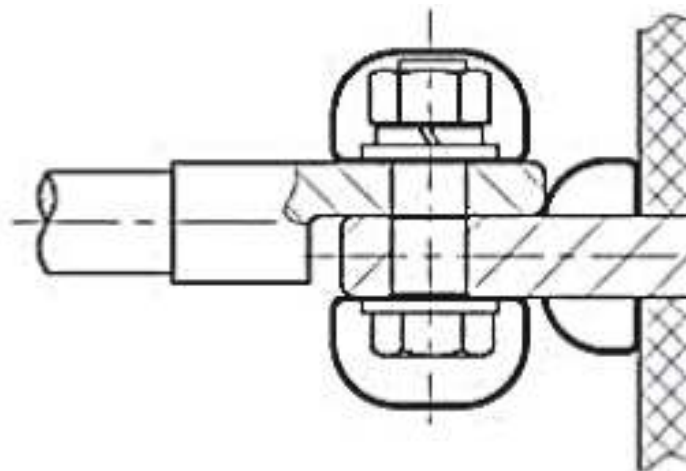


Figure 4.4.1

**DANGER!****Improper operation can cause damage to the tap-changer.**

Shielding caps supplied must be installed on the screws for 35KV and above.

The proper length of bolts must be used.

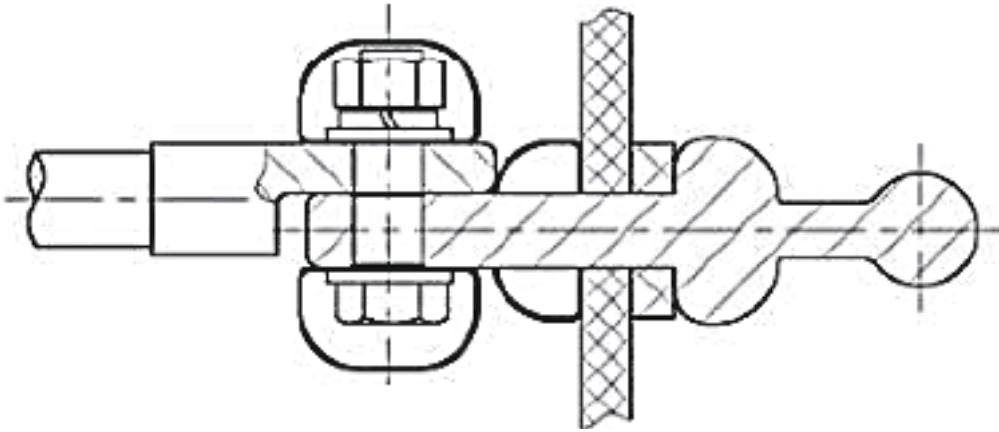


Figure 4.4.1

NOTE!**The tap selector terminal leads shall not deform or damage the tap selector under stress.**

- The leads of the transformer regulator winding shall be led to the tap selector from both sides respectively, to avoid unilateral lead force, resulting in distortion of the tap selector.
- The connection between the tap selector terminal and the terminal clamp of the transformer lead should have a certain deflection, which should not be too short, the connecting wire should be soft, it is recommended that the section of the connecting wire is not coated with insulating varnish, to avoid hardening and deformation of the insulating rod after drying.
- The end of the leads connecting the tap selector shall be connected in the shape of expanding ring (looped), so that the tap selector insulation board is not subjected to tension.
- Tap selector terminal leads should be led out from the outside of the selector and shall never pass through the inside of the selector.
- The bell-type on-load tap-changer must be lifted by 5 to 20 mm after the leads are connected. For this reason, the tightness of the connecting leads should be paid special attention. It is suggested that the spacer flange should be mounted on a support structure, that the spacer flange should be temporarily raised up to the final mounting and then the leads are led (same as real mounting). After the completion of the lead, remove the temporary support, check the tightness of the lead and whether the tap-changer is stressed.



5. Transformer Ratio Test and DC Circuit Resistance Measurement

It is recommended to carry out ratio test and DC resistance measurement prior to drying treatment.

NOTE!



Operation without oil will cause damage to tap changer.

More than 10 cycles operation without oil is not allowed.

For the first operation of tap -changer after transformer being dried, make sure tap changer is fully immersed with oil!

Incomplete tap changer operation will cause damage of the tap changer!

5.1 Ratio test (upper head flange is required for bell type tap changer)



NOTE!

For bell-type tap-changer, head flange, selector switch and top cover need to be assembled temporarily. (please refer to Fig 5.1.1)

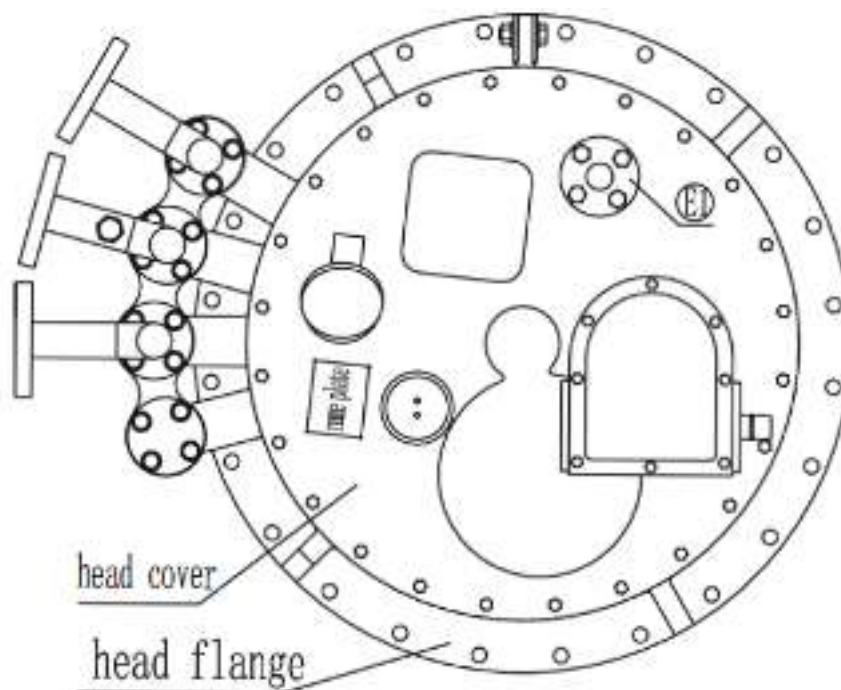


Figure 5.1.1

- 5.1.1 Turn the drive shaft of the tap changer top gear box, a short tube with an inner diameter of 25mm (a diameter of 12mm) can be used. Connection bolts have been screwed into the tube), and then equipped with a handwheel or hand crank.
- 5.1.2 For a group of two or three connected tap changers, all tap changer heads should be connected to each other with horizontal shaft
- 5.1.3 For one tap change, the sound of the selector switch action is clearly audible. After the sound of the selector switch be heard, it must continue to rotate two and a half turns in the same direction to complete the first stage tap-change.
- 5.1.4 In the process of operation, each operation position should be observed at any time through the observation window on the tap changer head to avoid exceeding the terminal position of the tap changer.

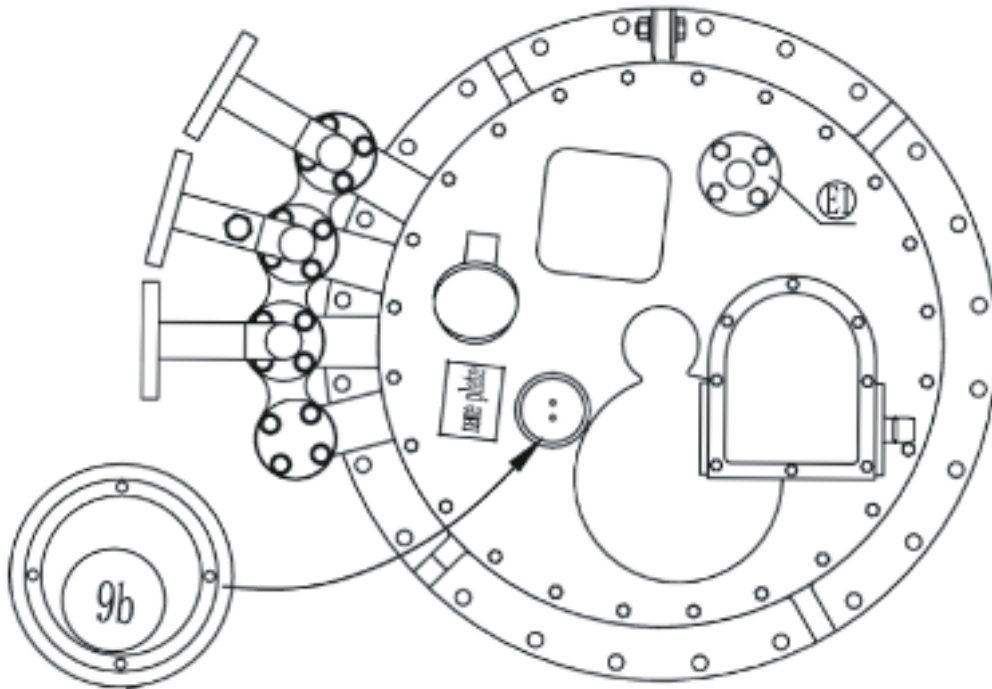


Figure 5.1.2

5.2 Transformer DC circuit resistance measurement

Kindly note that the tap changer is operated without oil when performing transformer DC resistance measurements, minimizing the operation of the tap changer is highly recommended.

After the transformer ratio test and DC resistance measurement, the tap changer must be adjusted to the set position.

6. Dry Processing and Oiling

6.1 Dry processing

In order to guarantee the insulation performance of tap changer, the necessary drying must be carried out according to the following instructions (vacuum drying or vapor drying).



NOTE!

Below parts, which are not allowed to be dried, they must be removed before the drying process: tap-changer top cover, top gear box, motor drive unit, protective relay, pressure relief device, bevel gear box, etc.

6.1.1 Vacuum drying

6.1.1.1 Drying in the oven room

- a. The tap-changer is heated up at a rate of 10°C/ hour in the air under atmospheric pressure, till the temperature reaches highest 110°C.
- b. Drying is carried out in the circulating hot air for 20 hours and highest temperature of tap-changer is 110°C.
- c. Vacuum drying lasts 50 hours, temperature of tap-changer is 105°C till 125°C. Maximum residual pressure 133Pa.

6.1.1.2 Drying in the transformer oil tank

- a. By-pass pipe (nominal diameter 25mm) can be connected to the flange connection position between flanges E2 and Q.
- b. The tap-changer is heated up at a rate of 10°C/ hour in the air under atmospheric pressure, till the temperature reaches highest 110°C.
- c. Drying is carried out in the circulating hot air for 20 hours under max. tap-changer temperature 110°C.
- d. Vacuum drying lasts 50 hours, temperature of tap-changer is 105°C till 110°C. Maximum residual pressure 133Pa.

6.1.2 Vapor phase drying



CAUTION!

OLTC top cover, top gear box and OLTC accessories are not allowed to be vacuum dried together with the transformer in the drying oven. Otherwise, they will be damaged.

6.1.2.1 Vapor phase drying in the vacuum tank



NOTE!

Top cover, top gear box and accessories of the on-load tap-changer are not allowed to be dried in the vacuum tank together with the transformer.

Before drying process, below parts must be removed and placed outside the drying oven, tap-changer top cover, top gear box, motor drive unit, protective relay, pressure relief device, bevel gear box, temperature sensor, etc. If it is vapor phase drying, be sure to open the oil drain bolt at the bottom of the oil compartment before drying, so that the condensate can be discharged from the oil compartment.

6.1.2.1.1 Loosen oil drain stopper

Use a special wrench to loosen the oil drain bolt counterclockwise. Complete loosening of oil drain bolt is not allowed.

- Turn clockwise to loosen the oil drain bolt on the oil compartment bottom. (See Figure 6.1.1)
- Remove the tap-changer top cover and store it outside the vacuum oven.
- Remove all accessories except the tap-changer body and put them outside the drying oven.
- Pass in kerosene vapor around 90°C and keep this temperature constant for 3 to 4 hours.
- Increase the kerosene gas temperature at a rate of 10°C/hour till the specified final temperature, but maximum temperature of the tap-changer cannot exceed 125°C.
- In vacuum drying, the temperature of the tap-changer is 105°C to a maximum of 125°C. The highest residual pressure is 133Pa. The drying duration is the same as the transformer, but at least 50 hours.
- Close the oil drain bolt (torque 18~20N.m.) after drying process.

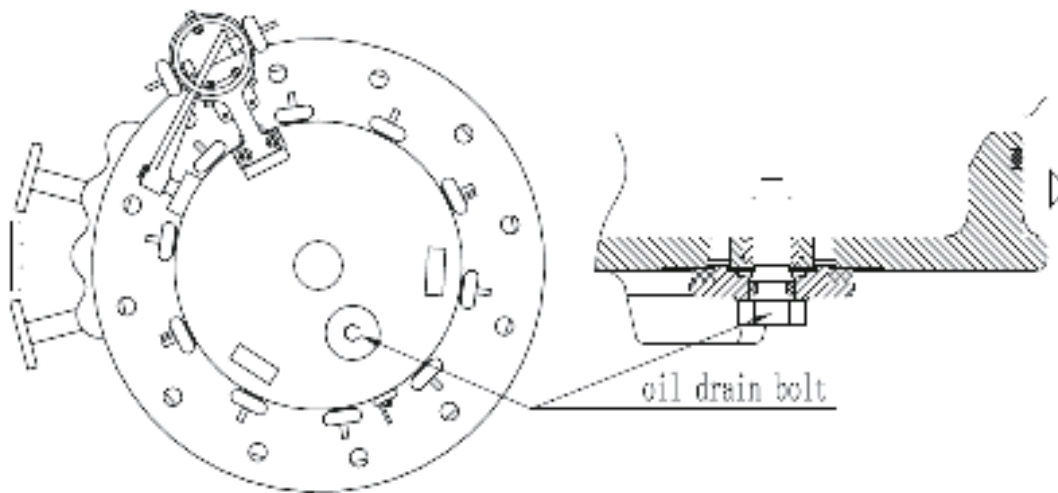


Figure 6.1.1



NOTE!

Use a special wrench to loosen the oil drain bolt counterclockwise, the oil drain bolt should not be loosened more than 10 cycles.

6.1.2.2 Vapor phase drying in the transformer tank

Before drying in the oil tank of the transformer, the selector switch insert should be lifted out, and the oil drain bolt at the bottom of the oil compartment should be loosened with a special wrench from in the oil compartment to discharge the condensate from the oil compartment.

6.1.2.2.1 Following items need to be paid attention.

- Make sure to prevent small parts from falling into the oil compartment.
- When disassembling and reassembling, the quantity of some small parts must be counted clearly.
- Avoid damaging the top cover of the on-load tap-changer and the sealing surface of the on-load tap-changer head during disassembly and other operations.
- Broken O-rings will cause the tank tightness to fail, resulting in oil leakage and damage to the tap-changer.
- Oil leakage will damage OLTC and transformer damage! Loose oil drain stopper will cause oil to leak from the oil compartment which will damage OLTC and transformer.

6.1.2.2.2 Main points for operation

- Avoid damaging the tap changer head cover and the sealing surfaces of tap changer head during disassembly and other operations.
- Before operating the tap changer for the first time after the transformer drying, make sure that the tap changer is completely immersed in transformer oil.
- If the tap changer is drying in the transformer tank, loosen the drain plug.
- After drying, do not close the oil drain bolt in the oil chamber, close the plug.

6.1.2.2.3 Pass kerosene vapor into on-load tap-changer

The pipe joints R and Q on the tap-changer head (See Figure-6.1.2) for the position of the pipe joint on the tap-changer head) are connected to the connecting pipe, and the inner diameter is at least 50mm to connect to kerosene vapor.

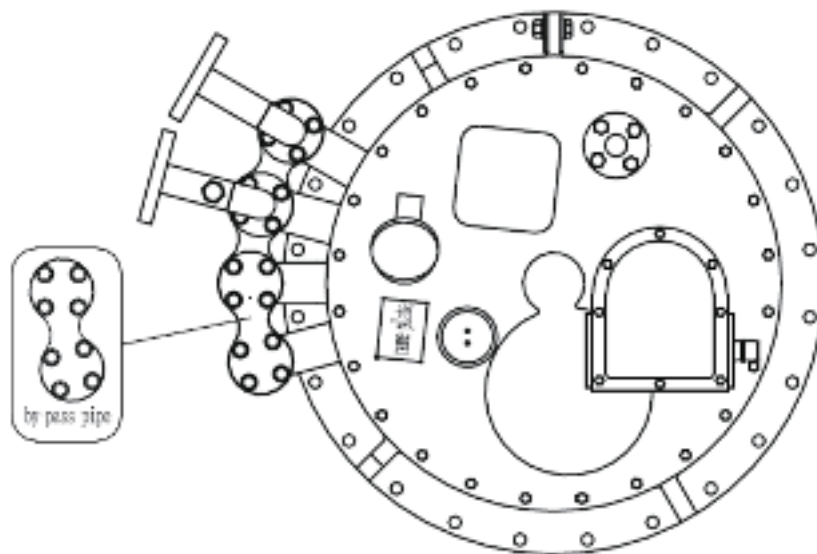


Figure 6.1.2

6.1.2.2.2 Drying

- a. Pass in kerosene vapor around 90°C and keep this temperature constant for 3 to 4 hours.
- b. Increase the kerosene gas temperature at a rate of 10°C/hour till the specified final temperature, but the maximum temperature of the tap-changer cannot exceed 125°C.
- c. In vacuum drying, the temperature on the tap-changer is 105°C to a maximum of 125°C. The highest residual pressure is 133Pa. The drying duration is the same as the transformer, but at least 50 hours.

6.2 Oil filling

After drying, in order to prevent too much moisture from entering the oil compartment, the oil compartment (selector switch insert already installed) should be refilled with oil as soon as possible.

The top cover of the tap-changer should be closed again and 24 pieces of M10 bolts should be tightened, paying attention to the correct position of the sealing O-ring. Both the transformer and the selector switch are filled with oil under vacuum. In order to vacuumize the selector oil compartment and the transformer simultaneously, the by-pass pipe accessories provided by our factory can be installed between oil filling flange on the tap-changer head and oil-overflow flange of the transformer.

Fill the tap-changer with new transformer oil up to the same level as in the transformer tank. Thus, install the by-pass pipe provided by our factory between the oil filling flange of the tap-changer head and the oil-overflow flange of the transformer, so that the selector switch oil compartment and the transformer can be vacuumized simultaneously.

- 1) In order to vacuumize the tap changer and the transformer at the same time, connect a by-pass pipe and the appropriate valves between pipe joints E2 and Q.
- 2) Both the on-load tap-changer and the transformer should be filled with new oil under vacuum at the same time.
- 3) Only new transformer oil with standard is allowed to fill for oil compartment and conservator.

7. Installation of Tap-changer Protection Device and Drive Unit Components

7.1 Installation of protective relay

Install according to the instruction manual provided when the protective relay is supplied.

7.2 Use of pressure relief device

The red bar protective plate on the top of the pressure relief device must be removed before operation (Figure 7.2.1).

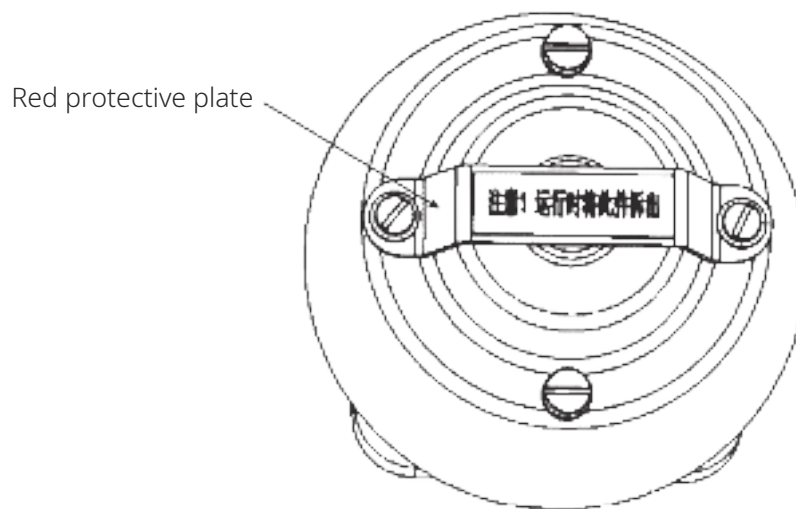


Figure 7.2.1

7.3 Installation of motor drive unit (MDU)

The motor drive unit (MDU) is used to drive the tap-changer to carry out the tap change operation; it can be operated automatically or manually.

Attentions should be paid when installing the MDU:

- 7.3.1 The MDU and tap-changer must be in the same setting position, which is indicated in the tap-changer connection diagram supplied with the equipment.
- 7.3.2 The MDU must be mounted at the provided place on the transformer tank in a vertical position, deflection is not allowed;
- 7.3.3 The installation plate for the motor-drive unit must be flat to prevent distortion and deformation of the motor mechanism, which could affect its functionality.
- 7.3.4 For detailed installation instructions regarding the motor mechanism, please refer to the corresponding user manual provided by Huaming Company.

7.4 Installation of the Bevel Gearbox

The bevel gearbox is secured to the bracket on the transformer tank cover with two bolts (hole diameter 18mm).

NOTE!

- The serial number marked on the nameplate of the bevel gear box must be consistent with the serial number on the nameplate of the tap-changer.
- When horizontal and vertical driving shafts are over two meters, an intermediate support gear box must be added. The installation of the specially designed bevel gear box, steering gear box and the intermediate support gear box of the vertical or horizontal driving shaft can be carried out according to the above instructions.

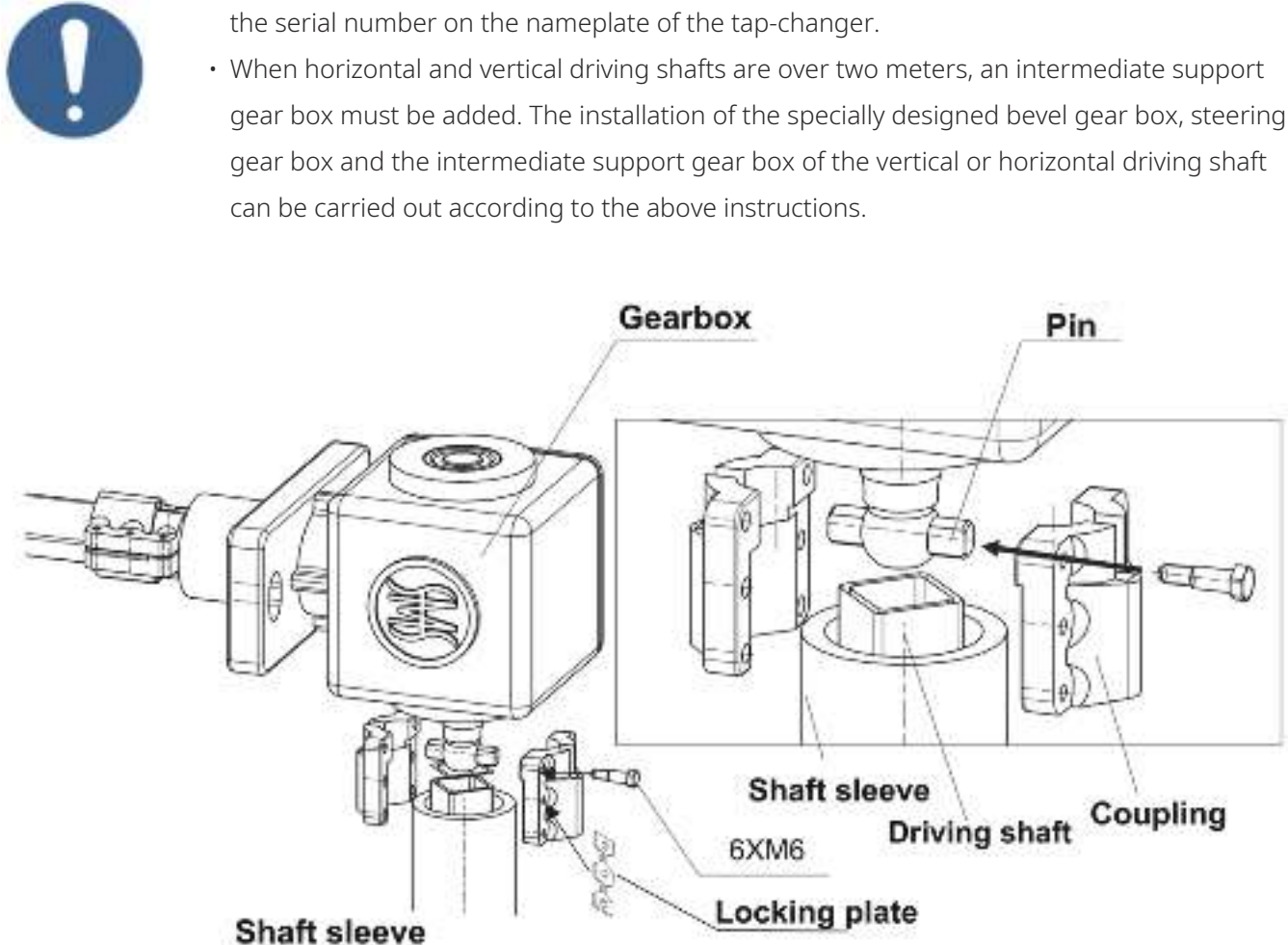


Figure 7.4.1

7.5 Installation of horizontal and vertical drive shafts

7.5.1 Connect the drive shafts, motor-drive unit, and on-load tap-changer as per the instructions provided in the Huaming motor-drive unit user manual (Figure 7.5.1 Appendix 3)

7.5.2. Mid-support box should be added if the vertical shaft length exceeds 2M. (See Figure-7.5.2)

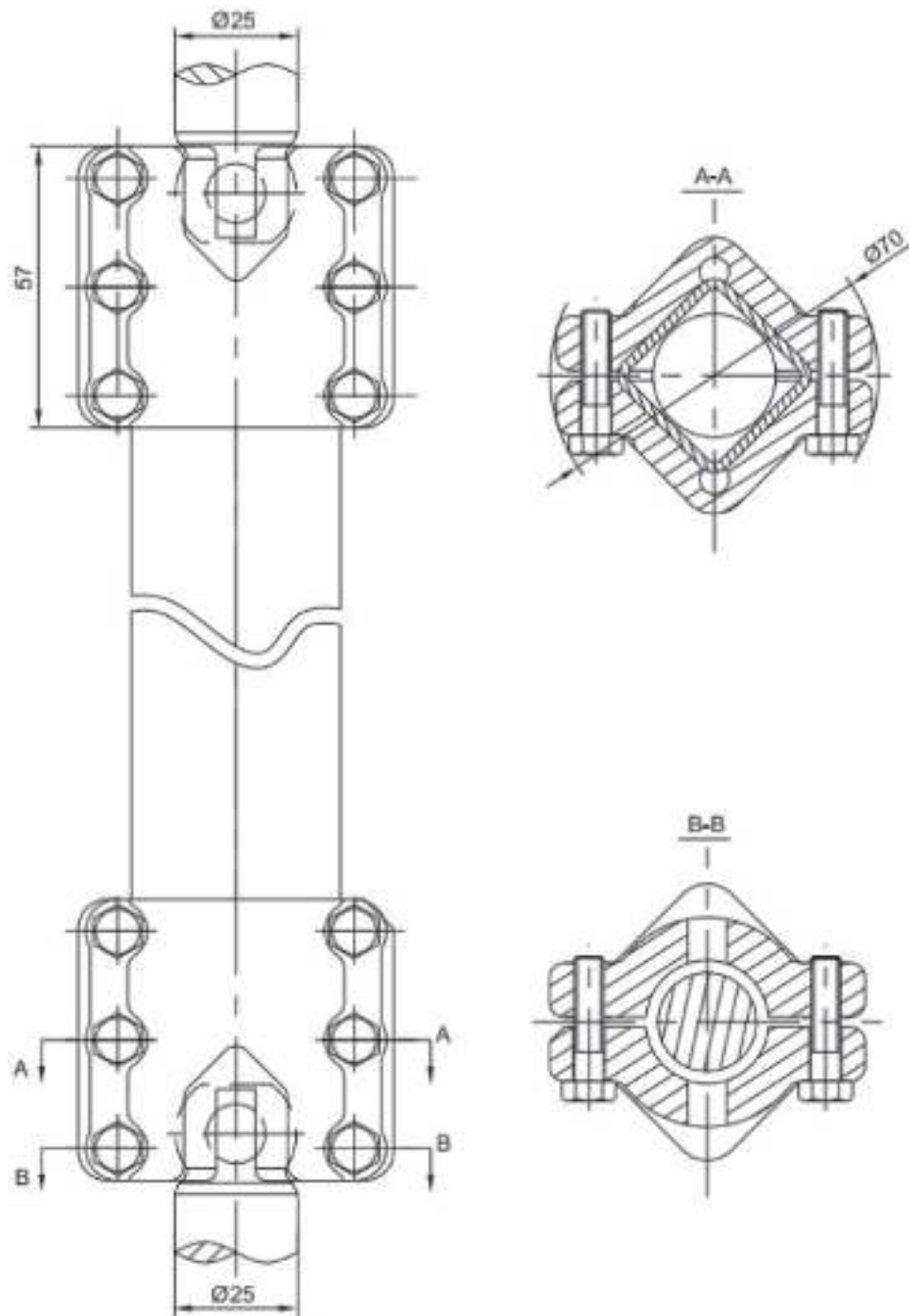


Figure 7.5.1 unit mm

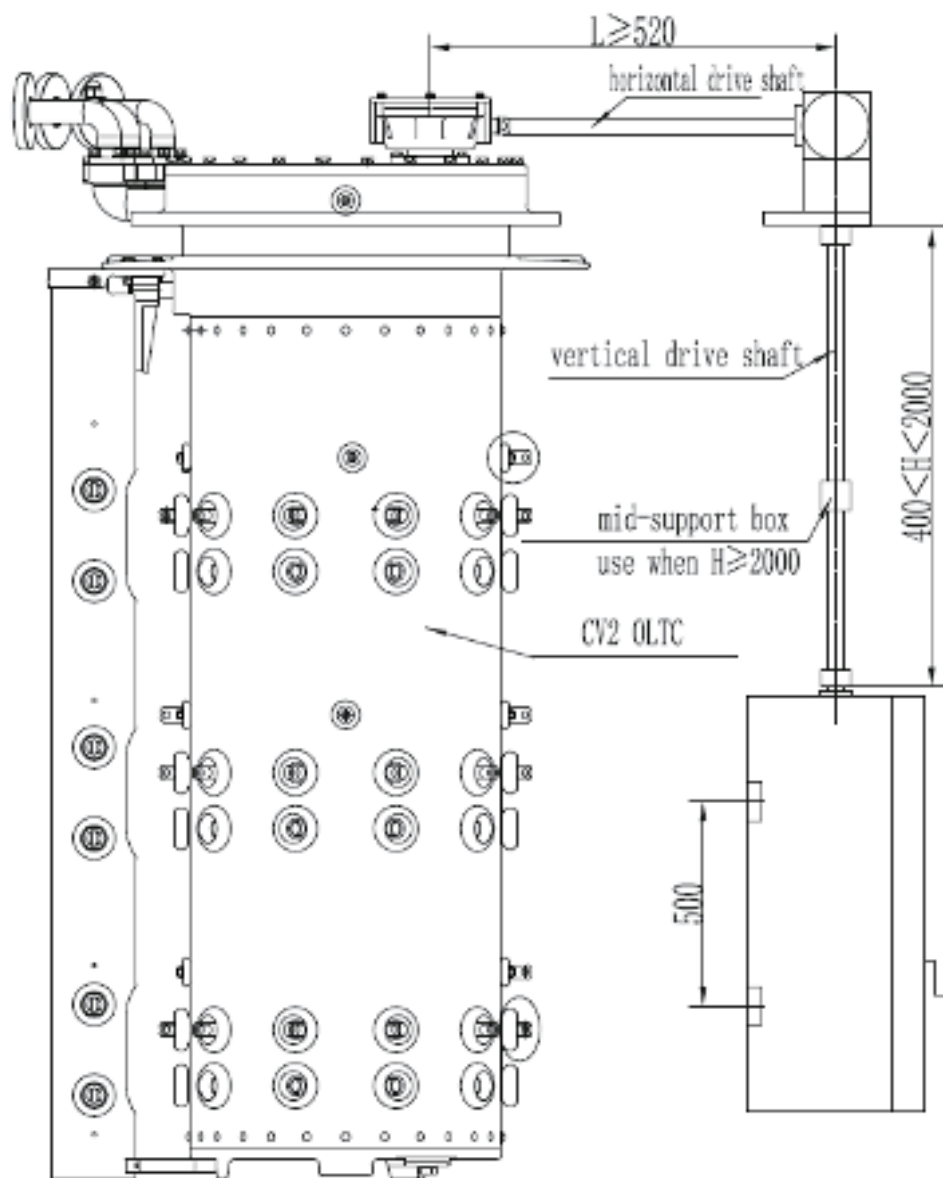


Figure 7.5.2 Appendix 5&6 unit: mm

7.6 Installation of driving shaft protective cover

7.6.1 Installation of horizontal protective cover

The length of the horizontal protective cover is determined by the transformer manufacturer's design. After the installation of the drive shaft is completed, place clamps (provided in the accessories) on both ends of the horizontal protective cover. Then, secure the cover by clamping it onto the protrusions of the conical gear case and bevel gearbox. Finally, move the clamps to the positions on both sides of the protrusions, tightening them securely (refer to Figure 7.6.1).

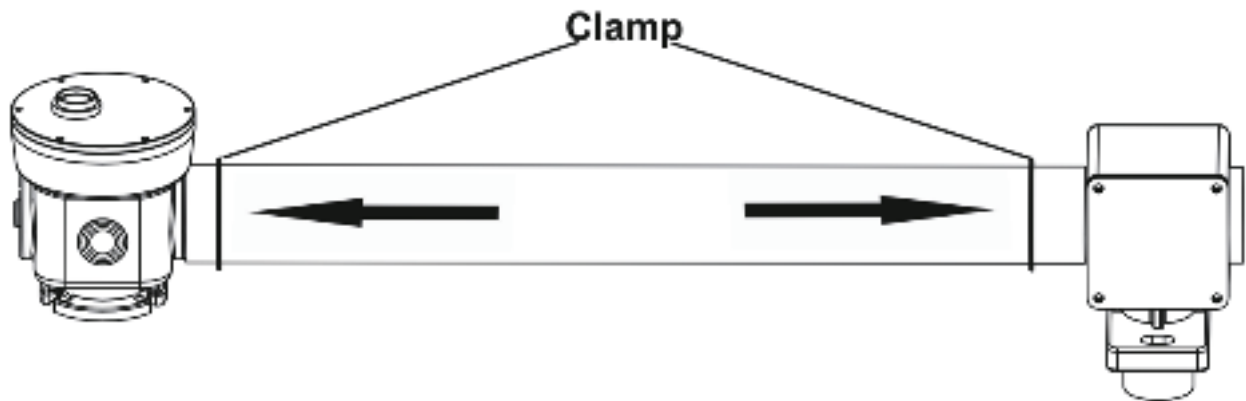


Figure 7.6.1

7.6.2 Installation of the Vertical Protective Cover

- 1) Secure three stainless steel cylindrical pins (6×16) onto the bushing (equally spaced at 120° intervals), ensuring they protrude approximately 5mm beyond the outer diameter of the bushing while not extending into the inner diameter.
- 2) Place the bushing at the lower end of the bevel gearbox and secure it with four M6×10 locking screws (equally spaced at 90° intervals) (refer to Figure 7.6.2).

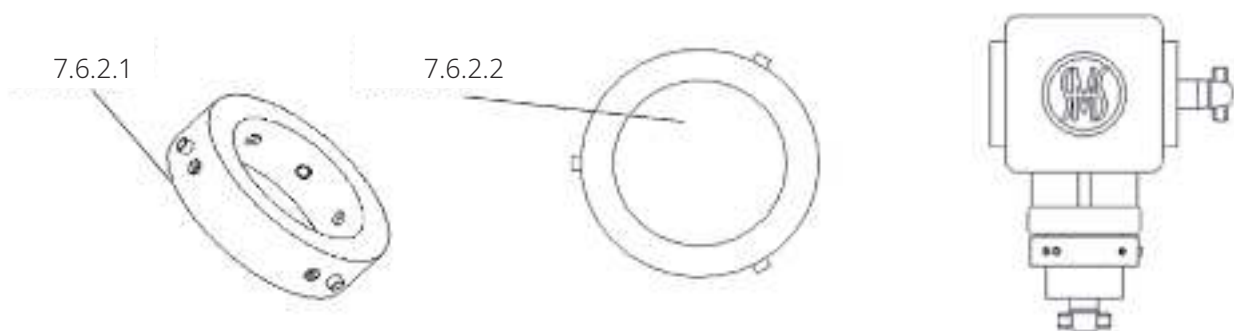


Figure 7.6.2

**NOTE!**

When the length of the vertical drive shaft exceeds 2 meters, an intermediate Connecting shaft is required.

- 3) Place the bushing at the upper end of the motor mechanism and secure it with four M6×10 locking screws, ensuring that the four screws are positioned to avoid the circular arcs (refer to Figure 7.6.3).
- 4) Slip both protective covers onto the outer side of the drive shaft, with notches distributed at both ends. The thicker one (HM8170.301) should be positioned above. Then, connect them together between the bevel gearbox and the motor mechanism following the pattern shown on the right side of the diagram.
- 5) Slip the lower protective cover onto the outer side of the bushing, ensuring that the four notches are positioned to avoid the four screws (refer to Figure 7.6.3).

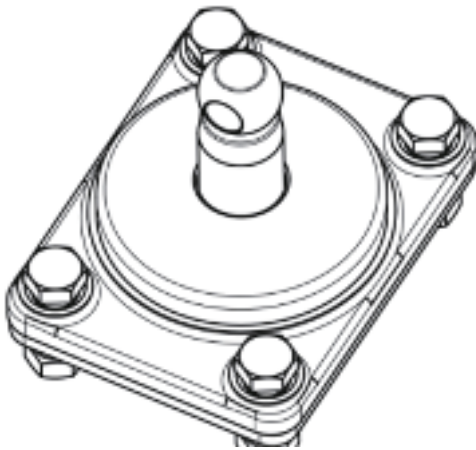


Figure 7.6.3



Figure 7.6.4

- 6) Pull the protective cover up to the lower end of the bevel gearbox, fitting it over the bushing, with the three notches aligned with the three cylindrical pins. Then, rotate the protective cover so that the cylindrical pins securely engage with the horizontal grooves (refer to Figure 7.6.5). (Note: The three pins are for temporary suspension only.)
- 7) At this point, both protective covers are properly positioned. On the lower end of the thinner protective cover, secure it to the upper end of the thicker protective cover using two clamps.



Figure 7.6.5

7.7 Installation of three single phase Tap changer and I+II tap changer



In the process of interconnecting three single-phase tap changers and I+II tap changers, below points should be taken into consideration:

All three single-phase OLTC and the I+II OLTC are located on the same transformer. since it will cause the operation of the selector switch during the adjustment of gear box, so after adjustment, must check if each selector switch is on same position and they have simultaneous action when operated by hand crank.

Attention: the arrangement of three single phase tap changers on one transformer, since it will cause the operation of the selector switch during the adjustment of gear box, so after adjustment, must check if each selector switch is on same position and they have simultaneous action when operated by hand crank.

7.8 Calibration of tap-changer transmission system

Manual operation for one cycle before electric operation is required when the tap changer is connected to the electric mechanism

The time interval between the moment the switch is switched to the completion of the MDU(Motor drive unit) is required to be the same for the two rotation directions When the tap changer is connected to the MDU.

The connection verification of the tap changer and the MDU normally has been performed before the factory test. But in order to ensure the reliability of the tap changer, below procedure shall be applied for the connection verification.

- 1) Rotate in the 1->N direction. When the switch is activated (when you hear the switch sound, continue to turn the handle and record the number of rotations, until the red center mark in the green band on the tap change operation indicator wheel of the electric mechanism appears in the middle of the observation window. Stop shaking at the time, and record the number of rotation m.
- 2) Shake the handle in the opposite direction N->1 to return to the original setting position, and also record the number of rotation K (follow item 1 instruction).
- 3) If the number of rotations $m = k$, the connection is correct. If $m \neq k$ and $m - k > 1$, the balance of the rotation difference is required. Loosen the connection of the vertical shaft of the electric mechanism, shake $(m-k) 1/2$ lap in the direction of multiple turns, and finally connect the vertical drive shaft with the electric mechanism.
- 4) Following the above steps, check the rotation difference between the electric mechanism and the tap changer, until the correction results in the same number of turns, that is, $m=k$

For example:

The connection verification between the selector 10193W and the CMA7 MDU: from the 10 positions (setting position) to the 11 position, $m = 5$ circle, from the 11 position back to the 10 position (the original setting position) $k = 3$ circle, the handle rotates a circle Number difference: $mk = 5 - 3 = 2$ turns

Adjust the number of turns (mk) $1/2 = 1/2 (5 - 3) = 1$ turn. Loosen the connection between the vertical drive shaft and the electric mechanism and turn the handle to position 10 -> position 11 according to the number of adjustments turns 24 above. Then connect again.

Check the balance of the number of rotational differences between the two directions of rotation.

- 1) Record the number of m and k turns under the connected condition.
- 2) Disconnect and shake in most directions $(m-k)1/2$
- 3) Re-connect and verify to $m=k$ turns

7.9 Procedure of take out the insert of Tap Changer

Usually, the tap changer is stopped at a fixed position and the insert is suspended. The specific position is shown in Table 7.10

Type of tap changer	Position
10040, 10050...10100, 10191W/G, 10193W/G	Position 3
10071W	Position 7
10091W	Position 8
12110, 12120, 12231W/G, 12233W/G	Position 4
12111W	Position 10

Table 7.10



Figure 7.10.1

a) Reduce oil level

Close the valve between the oil conservator and the tap changer, open the exhaust oil spill screw on the head cover, and then open the oil drain valve of the tap changer to lower the oil level of the tap changer.

b) Dismantling the head cover

Follow the steps in 4.2.1;

c) Disassemble the energy storage spring of the energy storage mechanism (Figure 7.10.1)

Take out three tension spring pins (item #2) and rotate the energy storage spring to separate it from the upper flange.

d) Remove the oil suction pipe (Figure 7.10.1)

- 1) Loosen the pipe nut (by No.38 wrench and refer to item #4) at the oil suction pipe;
- 2) Pull out the pipe connector (item #4);
- 3) Pull out the oil suction pipe (item #6).

e) Switch the moving contact of insert to the two fix contacts between of the oil compartment (refer to Figure 7.10.1)

- 1) Take out the red orientation pin (refer #1) on the flange surface and insert it into the red orientation pin hole on the mount (item #11);
- 2) Slowly rotate the driving gear (item #9) counterclockwise using the lever and shaft (item #6 and #7) until the orientation pin falls into the oil suction pipe (Figure 7.10.1); At this point, the clamp contact (moving contact of insert) is located between the two contacts of the oil compartment fix contacts.

f) Remove the fastening insert nut (Figure 7.10.1)

Disassemble the tap changer insert by using No.13 socket (item #3), Remove 5 pieces M8 nuts, and remove the nuts and their washers. Pay attention to keeping the nuts and washers well.

g) Pull out the central oil suction pipe (Figure 7.10.1)

- 1) Remove the 2 pieces M8 bolts (by using No.13 socket);
- 2) Take out the red semicircular plate (item #10);
- 3) Reassemble the 2 pieces M8 bolts (item #9);
- 4) Pull out the center oil pipe (item #13) by using the lever and shaft (item #6 and #7).

h) Slowly lift the tap changer core vertically (Figure 7.10.1) and place it in a flat and clean place.

7.10 Reassemble

Reinstallation is the reverse process of lifting out, and it must be carried out one by one according to the reverse process of lifting out, otherwise it will damage the equipment.

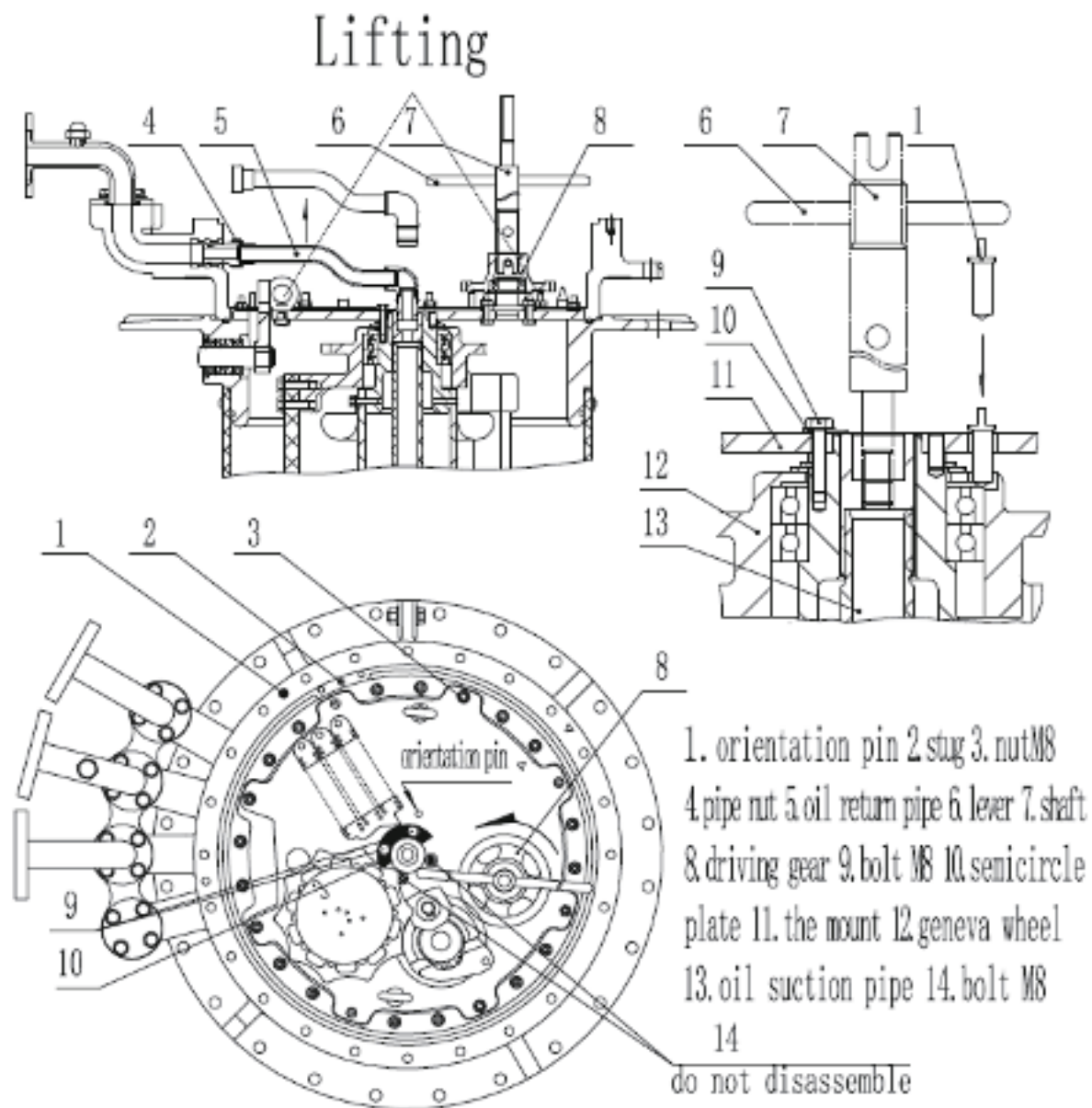


Figure 7.10.1



DANGER!

May cause equipment damage and personal injury.

When hanging the insert, the clamp moving contact must be between the two contacts of the oil compartment fix contact, otherwise it may damage the clamp moving contact or the oil compartment fix contact. (Refer to step e of chapter 7.9)

8. Tap-changer Factory Test and Test Preparation

8.1 Test preparation

8.1.1 OLTC Fully gas release

Before putting into operation for the first time, use the release plug of the suction pipe on the tap changer head to deflate the connection pipe.

8.1.1.1 Gas release for OLTC

- 1) Remove the M16 bolt cap on the release valve E1.
- 2) Pry up the valve stem with a slotted screwdriver to deflate the tap changer head until the oil flows out (Figure.8.1.1)
- 3) Tighten the bolt cap of the release valve E1 (torque 9~12 N.m.)

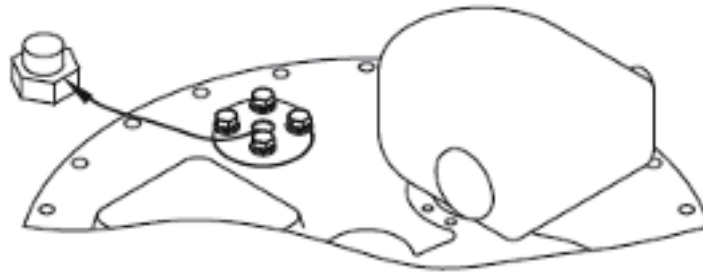
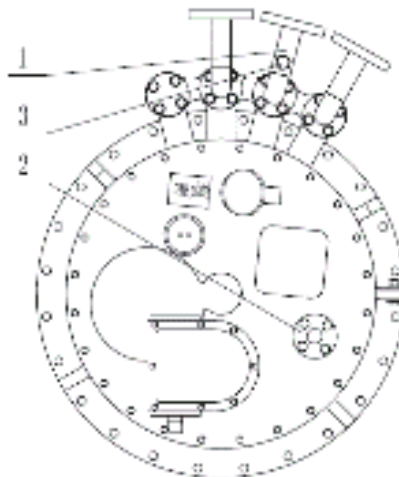


Figure 8.1.1

8.1.2 Gas release for the suction pipe on the pipe joint S

- 1) Remove the M16 bolt cap on the pipe joint S, and use a No. 16 wrench (Figure.8.1.2).
- 2) Open the notched bleed plug M6 connecting pipe to release gas;
- 3) Close the bleed plug (torque 2 N·m).
- 4) Close the bolt cap of the bleed plug. (torque 2 N·m).



1. pipe joint S 2. Bleed plug 3. M16 bolt cap

Figure 8.1.2

8.1.2 Grounding

8.1.2.1 Connect the ground bolt of the tap changer head to the transformer cover (1 piece of M12 bolt and nut), No. 19 wrench, torque 50~60 N·m, Figure.8.1.3.

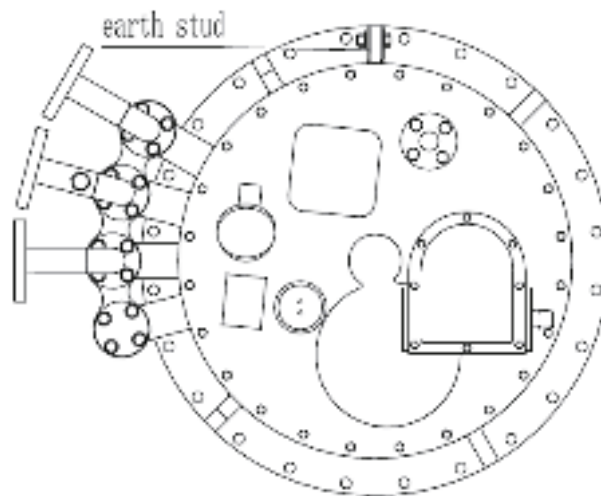


Figure 8.1.3

8.1.2.2 Connect the grounding bolt M12 (No. 19 wrench, torque 50~60 N·m) on the transmission mechanism protection box to the transformer oil tank.

8.2 Test in transformer factory

8.2.1 Operation test

Pre- operation must be carried out to check the mechanical functions of the tap-changer and the MDU before the transformer is energized.

Pay attention to the following operations during the test.

- 1) When the first time of operating the on-load tap-changer after the transformer has been dried, make sure that the tap changer oil compartment is filled with oil, operation without oil is forbidden.
- 2) During operation test, the tap changer must go through the complete operating range (full operation range, One complete cycle).
- 3) Make sure that the position indicators of the motor-drive mechanism and the tap changer (observation hole on the tap changer head) show the same position in each operating position.

WARNING!

Serious damage on equipment and accident may cause!

If the connection between the tap-changer and the MDU is improperly aligned, continuous operation will cause damage to the tap-changer.

Ensure that the position indicators of the MDU and tap changer are in the same position!



8.2.2 The electrical test of the transformer

WARNING!



May cause Serious damage to equipment, death or injury to personnel, and lead to further accidents

There is explosive gas under the tap-changer cover, in the connection system, oil conservator and the outlet of air dryer. The flying parts and the high temperature oil splashing to the outside will cause fatal and serious injury.

NOTE!



Please confirm: there is no open flame, heat source or electric spark (such as electric spark produced by static electricity) within its direct contact range, and there is no possibility of generating them. It is Ensured that the tap-changer test must be carried out by professionals in accordance with the test procedures.

9. Transformer Transportation and Put into Operation on Site

9.1 Remove the MDU for transportation

9.1.1 Place the MDU in the calibration position and then remove the coupling.

9.1.2 Remove the MDU.

9.1.3 Do not operate the MDU when the tap switch is not connected well. The reassembly of the MDU and drive shaft must be carried out according to Huaming's instructions.

9.2 Oil filling and transportation without oil conservator

If the transformer is filled with oil during storage and transportation but no oil conservator is installed, a bypass pipe must be installed between the inside of the selector switch oil chamber and the transformer oil tank to balance the pressure generated by oil expansion. The bypass tube is installed between the pipe connector E2 and Q.



NOTE!

For the storage period of 3 to 4 weeks without oil conservator, the oil level of the tap changer should be reduced by about 5 liters.

9.3 Transportation without oil

If the transformer is transported or stored without oil filling, all the oil in the tap-changer oil compartment must be discharged. The protection method for tap-changer oil compartment should be same as transformer, such as nitrogen filling.



NOTE!

For long-term storage, the heater of the MDU should be regularly energized to prevent moisture.

9.4 Putting transformer into operation on site



DANGER!

Can cause serious damage to equipment, death or injury to personnel, and lead to further accidents

Make sure there is no open flame, heat source or electric spark (such as electric spark produced by static electricity) within 50m range, and there is no possibility of generating it.

**NOTE!**

Operation without oil will lead to damage of OLTC.

Before putting the transformer into operation, make sure the tap selector is completely immersed in the transformer oil and as well as the oil compartment.

9.4.1 Operation on site:

- 9.4.1.1 When the transformer is put into operation for the first time Ud 40kV/2.5mm (Minimum), H₂O 12ppm (Maximum).
- 9.4.1.2 Gas release please follow OLTC chapter 8.1.
- 9.4.1.3 Pre-operation please follow chapter 8.2.
- 9.4.1.4 Please make sure that the lowest oil level signal contact in the oil conservator of the on-load tap-changer is connected to the trip circuit of the circuit breaker.
- 9.4.1.5 Check whether the gas relay function is normal according to the Huaming gas relay QJ-25 series instruction manual.
- 9.4.1.6 The red protective strip of the pressure relief valve must be removed before putting into operation.

It is necessary to make sure that the instantaneous impact of the current start is completely reduced before the tap change operation. Normally, the starting current is several times the transformer current, which may cause overload of the on-load tap-changer during switching, so it is recommended to wait for the current to return to normal half a minute after powering on before operating.

Observe the safety instructions described in all functional checks and tests during the commissioning.

After power is supplied to the transformer and the instantaneous impact of the current start is completely reduced, the tap change can be operated under no-load or under-load conditions.

10. Monitoring and Maintenance for Tap-changer Operation

10.1 Monitoring

To ensure proper operation of the on-load tap-changer, it is necessary to carry out regular visual inspections with the transformer.



CAUTION!

Inspection of OLTC head cover, protective relay and MDU should be carried out regularly.

Regulation type	Breakdown voltage	moisture content
Regulation in neutral point	$\geq 30 \text{ kV}/2.5 \text{ mm}$	$< 40 \text{ ppm}$
Except regulation in neutral point	$\geq 40 \text{ kV}/2.5 \text{ mm}$	$< 30 \text{ ppm}$

10.1.1 When the transformer is overloaded, OLTC cannot be operated excessively. "Overcurrent blocking" must be installed to prevent OLTC from being switched when the load current is more than 2 times I_n .

10.1.2 The trip contact of protective relay is set to operate when the oil speed is $1.0 \text{ m/s} \pm 10\%$. This contact should be connected to the trip circuit of the transformer's circuit breaker; When the fault occurs inside the OLTC, a large amount of gas will be generated, resulting in accelerated oil flow rate, so that the relay stops plate action, trip contact closed, to cut off the transformer power supply, to avoid the expansion of the accident. Once the gas relay is operated, it is strictly prohibited to switching the OLTC before maintenance.

10.1.3 Rupture disk is installed on the top of the tap switch, which will not move during the normal operation of the switch. The rupture will occur within the OLTC only if the pressure in the oil chamber exceeds $0.3 \pm 20\% \text{ MPa}$ and acts as a rupture protection to avoid further expansion of the accident. During OLTC installation and maintenance, be careful not to step on or hit the blasting cap with heavy objects.

11. Maintenance

Electric shock!

An energized transformer may cause death or serious injury.



- Turn off the high voltage and low voltage terminals of the transformer.
- Lock the transformer to prevent accidental restart.
- Ensure that all components are powered off.
- Ensure that all transformer terminals (earth wires, earth disconnect switches) are earthed and short-circuited.
- Cover or isolate neighboring live parts.

Electric shock!

Operation of energized on-load tap-changers may cause personnel death or serious injury.



- Disconnect all auxiliary circuits, such as tap change monitoring devices, pressure relief valves and pressure monitors.
- Ensure that all components are powered off.

Warning!

Risk of explosion!



- Explosive gas in the oil compartment of the on-load tap-changer, transformer, piping system, oil conservator, and the outlet of the dehydrating breather can ignite or explode, resulting in serious injury or death.
- Ensure that there are no open flames, hot surfaces or electrical sparks (e.g. caused by static build-up) in the environment surrounding the transformer.
- Do not operate any electrical equipment (for example, impact wrenches have the risk of sparking).
- Only use conductive and grounded hoses, pipes and pump operation equipment suitable for flammable liquids.

Check the MDU!

Condensation in the MDU can cause damage to the motor of the MDU.



- Always keep MDU's protection case sealed.
- If the operation is interrupted for more than 2 weeks, connect and operate the anti-condensation heater in the MDU. If this requirement cannot be met (for example, the heater cannot be energized during transportation), a sufficient amount of desiccant must be placed inside the case.

12. Inspection

12.1 Regular maintenance inspection

Please follow the inspection instruction below:

Interval	Instruction	Details
Once per year	Check the door sealing, cable bushings, MDU protective cabinet and control panel's ventilation device	Visually observe whether the sealing parts are aging, damaged, whether oil leakage, and if yes, replacement is required immediately. The ventilation of the MDU and control panel should be tested. If aging damage occurs, replacement is required immediately.
Once per year	Check the sealing of tap-changer top cover, protective relay and all pipes' connections.	Visually observe whether the sealing parts are aging, breakage, whether there is oil leakage, whether it is elastic, and if the seal is aged and damaged, replacement is required immediately.
Once per year	Check the MDU's protective cabinet and if the heater inside the control panel works properly.	Visually check whether the circuit components are black or yellow, the pins are broken, if there is damage, please contact Huaming service. Then use a thermometer to check whether the corresponding temperature has been reached.
Once per year	Check if the protective relay functions properly.	Visually check whether the circuit components are black or yellow, the pins are broken, if there is damage, please contact Huaming service.
Once per year	Check whether the silica gel desiccant in the transformer oil conservator is working properly	Put the humidity indicator card in the oil conservator for at least half an hour, if the humidity is greater than 10%, replace the desiccant, or replace the desiccant every other year.
Once per year or every 200,000 operations	Check the oil sample. If it doesn't comply with the parameters in Table 13-2, it's recommended to replace the transformer oil.	Take oil Samples 50-250 ml at the bottom using a syringe, if the parameters in Table 12.1.2 are not met, it is recommended to replace the transformer oil.

Table 12.1.1 Inspection instruction

Item		Breakdown voltage	Water content
Limit values for insulating oil in accordance with IEC 60296			
When commissioning the transformer for the first time		> 60kV/2.5mm	< 15ppm
During operation	Class I tap-changer	> 30kV/2.5mm	< 40ppm
	Class II tap-changer	> 40kV/2.5mm	< 30ppm
After maintenance		> 45kV/2.5mm	< 15ppm
Limit values for natural esters in accordance with IEC 62770			
When commissioning the transformer for the first time		> 60kV/2.5mm	≤ 100ppm
During operation	Class I tap-changer	> 30kV/2.5mm	≤ 200ppm
	Class II tap-changer	> 40kV/2.5mm	≤ 200ppm
After maintenance		> 45kV/2.5mm	≤ 100ppm

Table 12.1.2 Limit values for oil samples

12.2 Interval of Maintenance:

Interval	Instruction
After 300,000 times OLTC operation (based on the number shown in MDU's counter)	OLTC inspection and maintenance. Please contact Huaming's service department.
After 600,000 times OLTC operation (based on the number shown in MDU's counter,	Inspection for vacuum interrupters. Please contact Huaming's service department.
After 800,000 times OLTC operation (based on the number shown in MDU's counter)	Replace the tap changer insert. Please contact Huaming's service department.
After 1,500,000 times OLTC operation (based on the number shown in MDU's counter)	Tap selector maintenance. Please contact Huaming's service department.

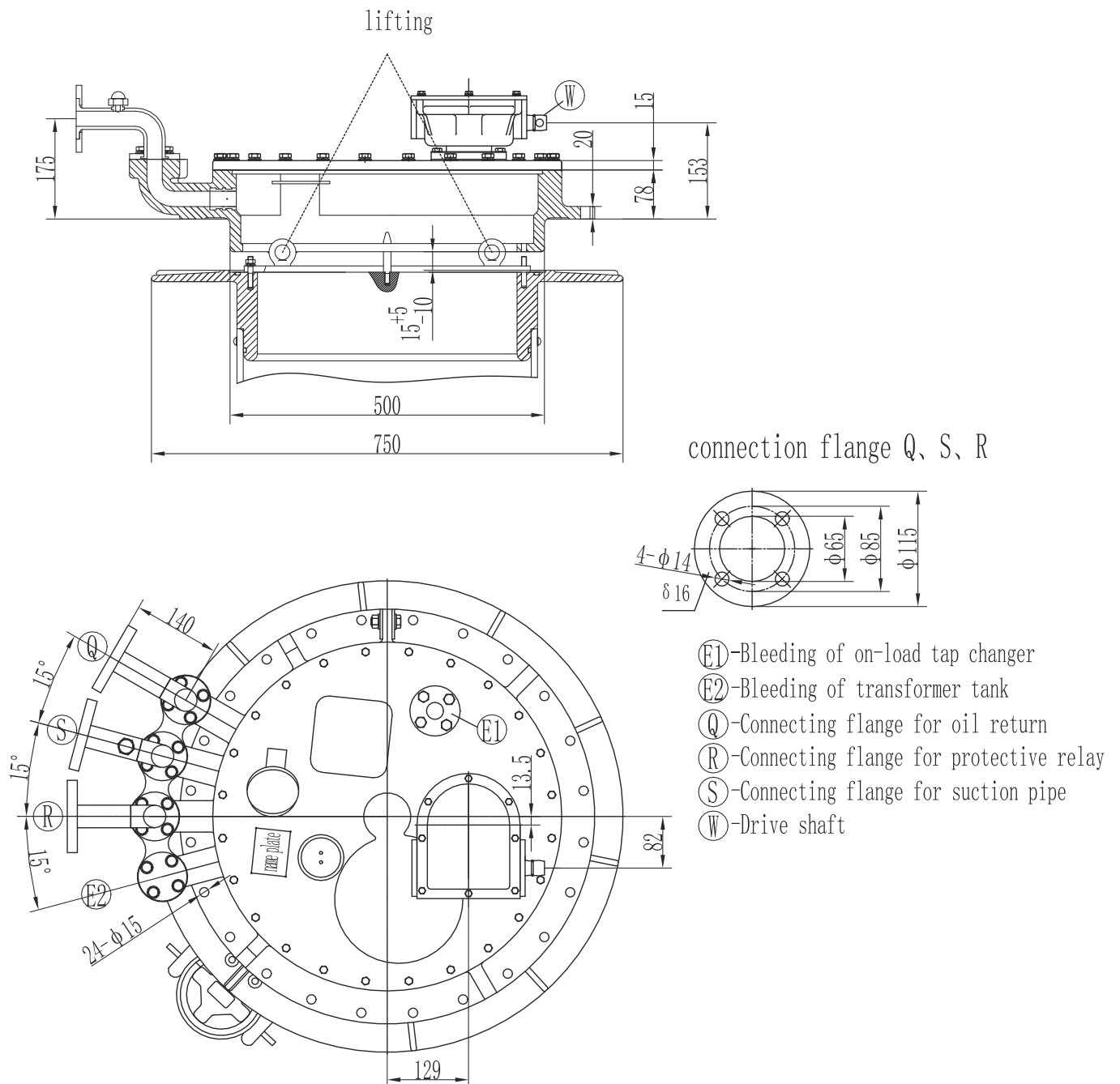
Table 12.1.3 Maintenance interval

Please write to export@huaming.com for any support you may need.

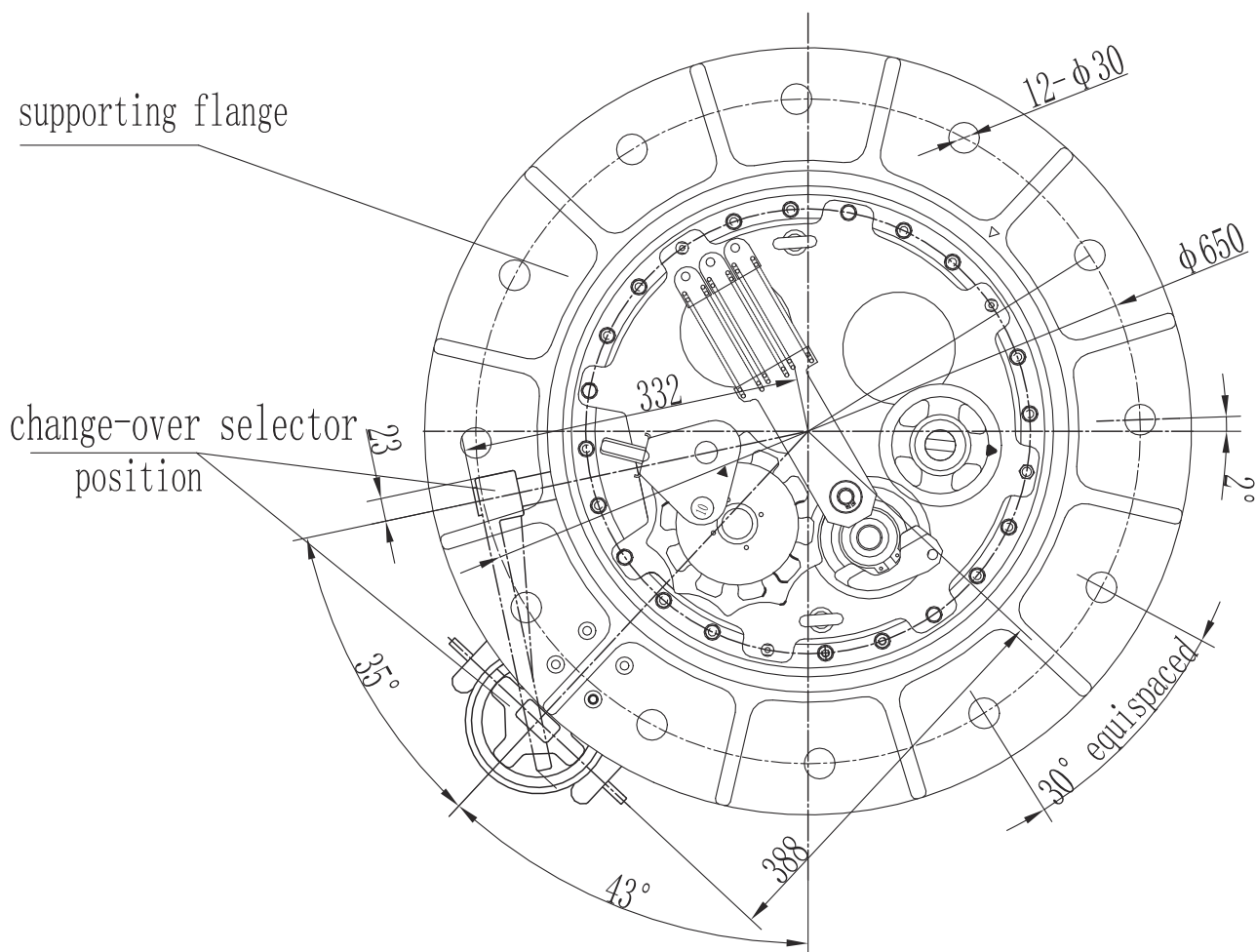
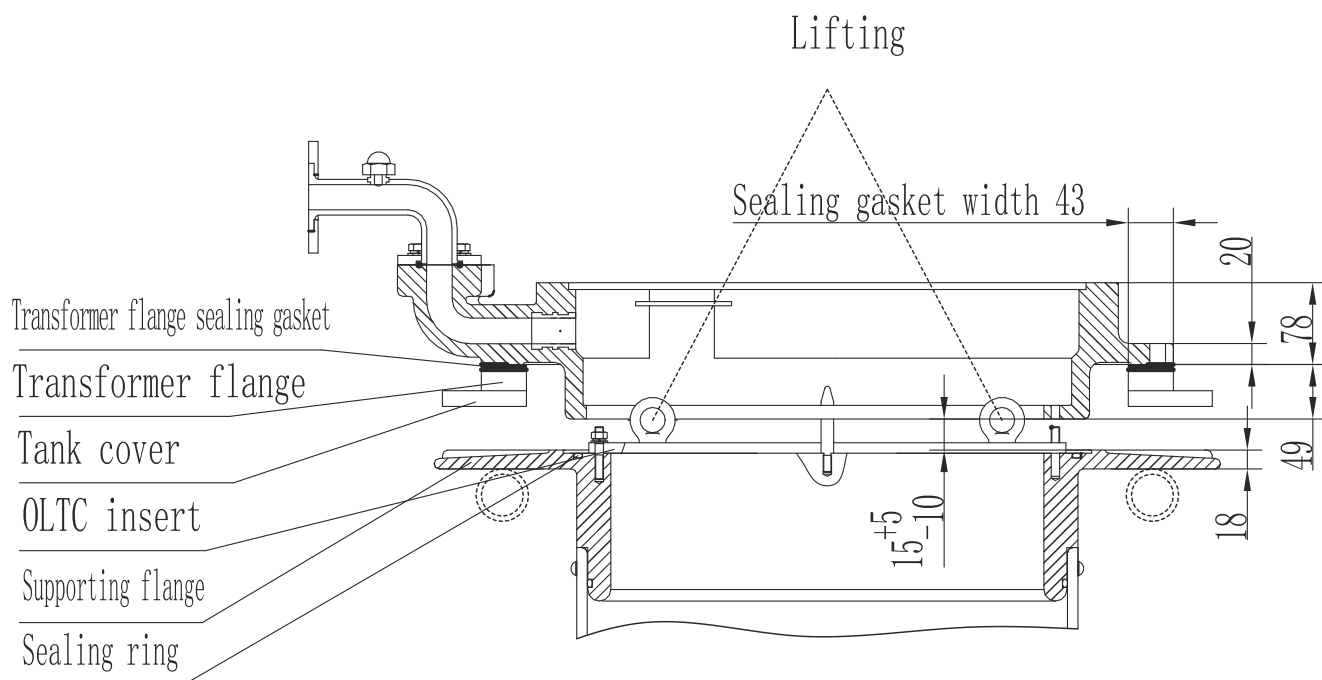
13. Appendix

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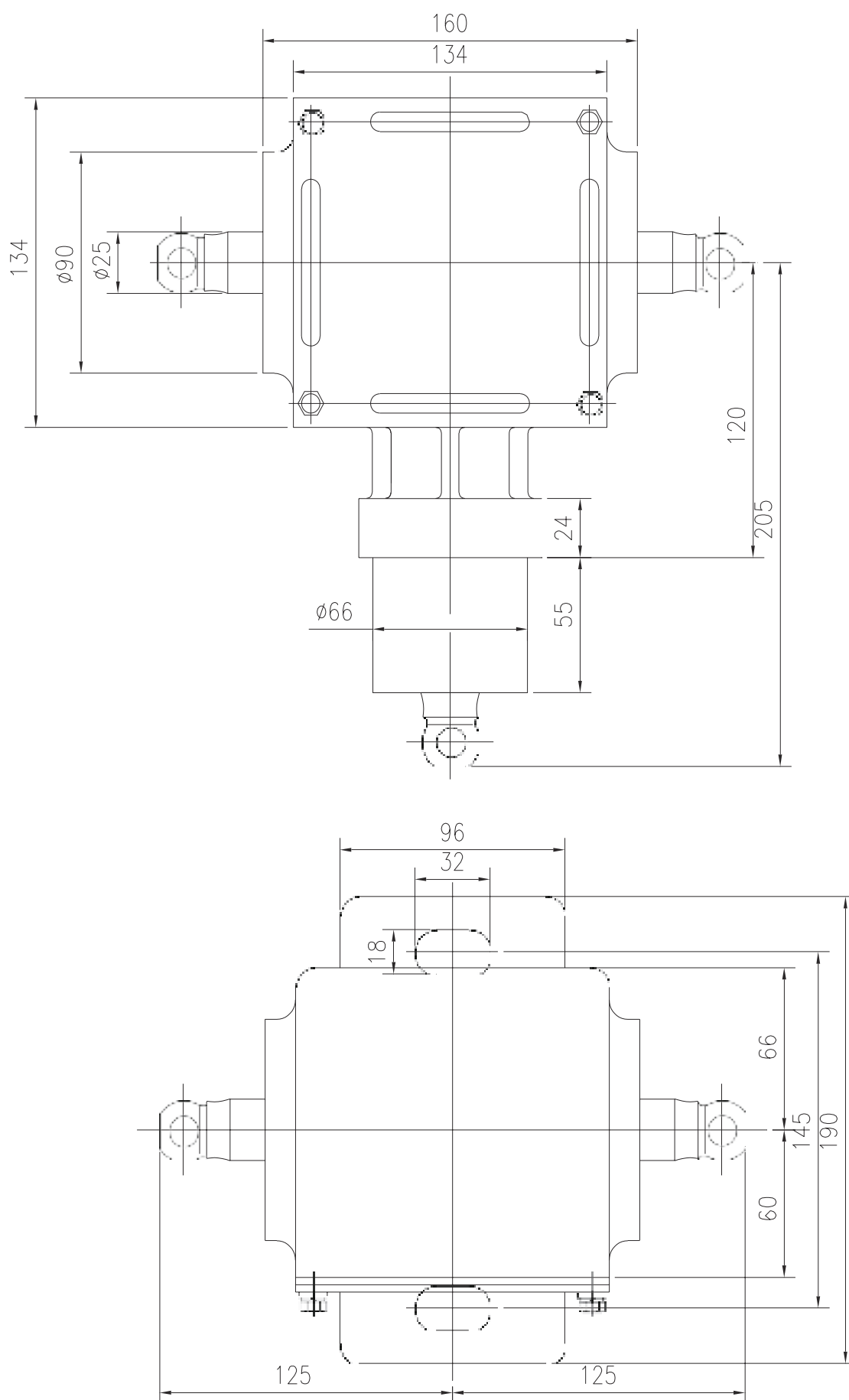
Appendix 1. CV2 head flange, overall dimension



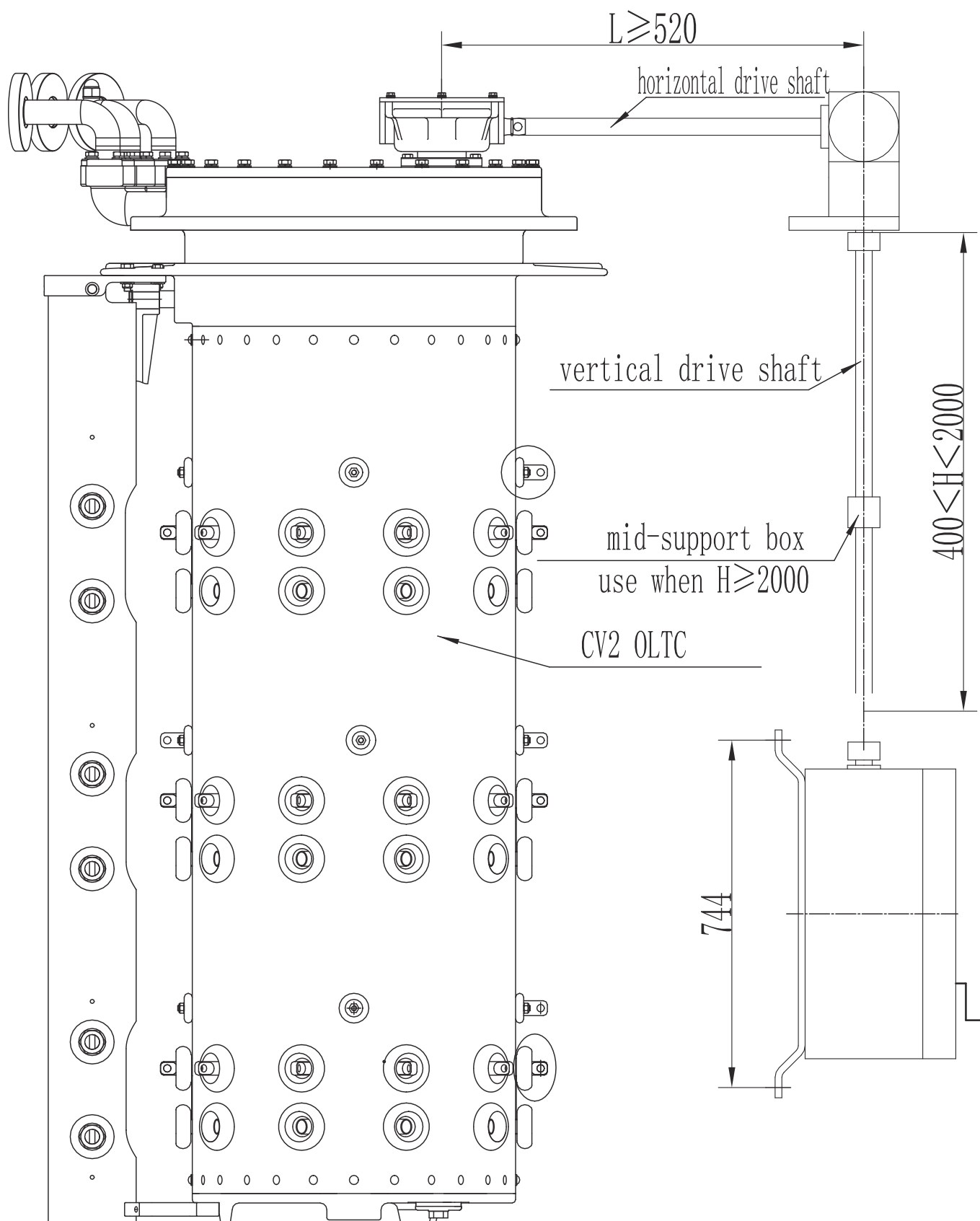
Appendix 2. The overall diemsnion of bell type



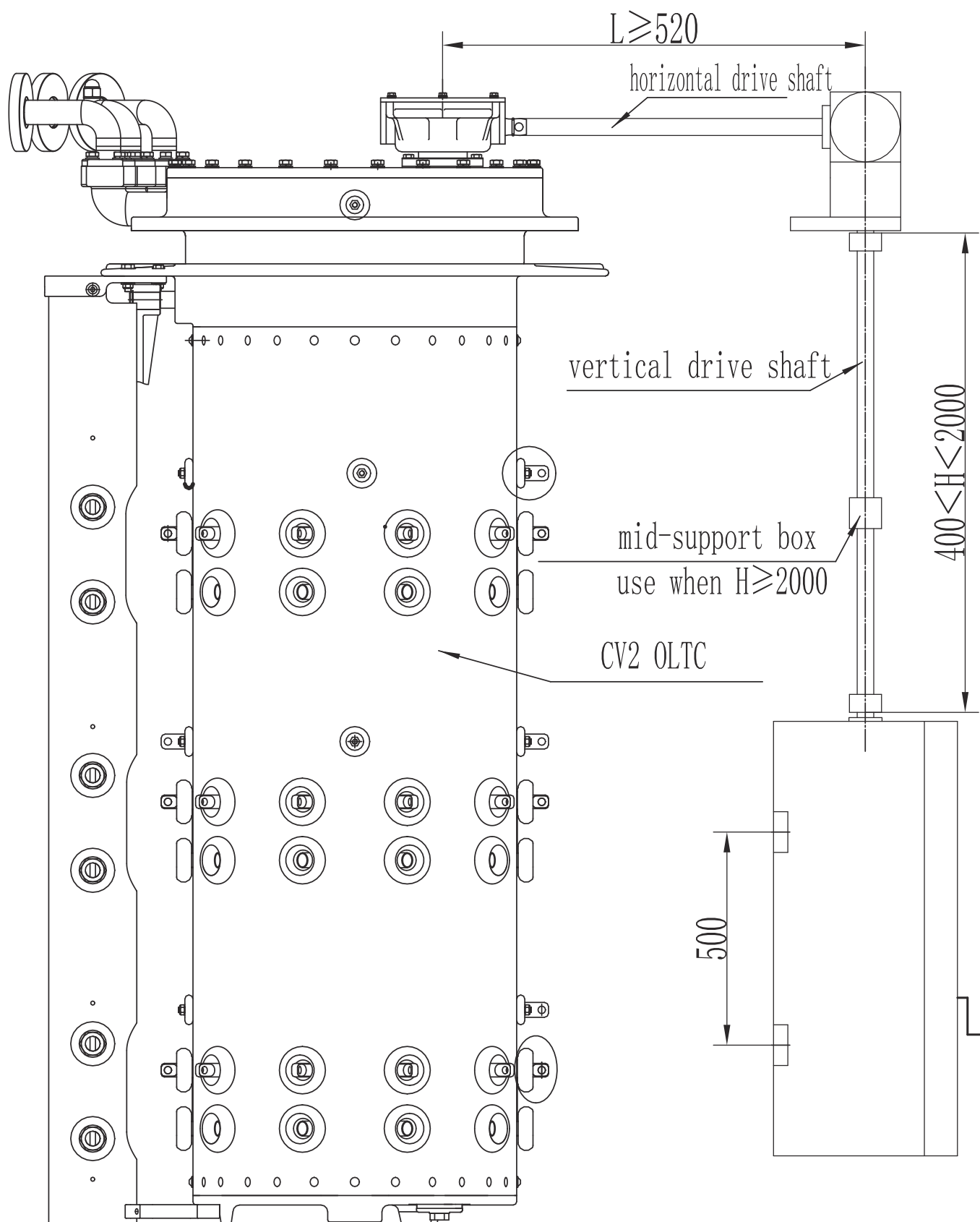
Appendix 3. The overall dimension of bevel gearbox

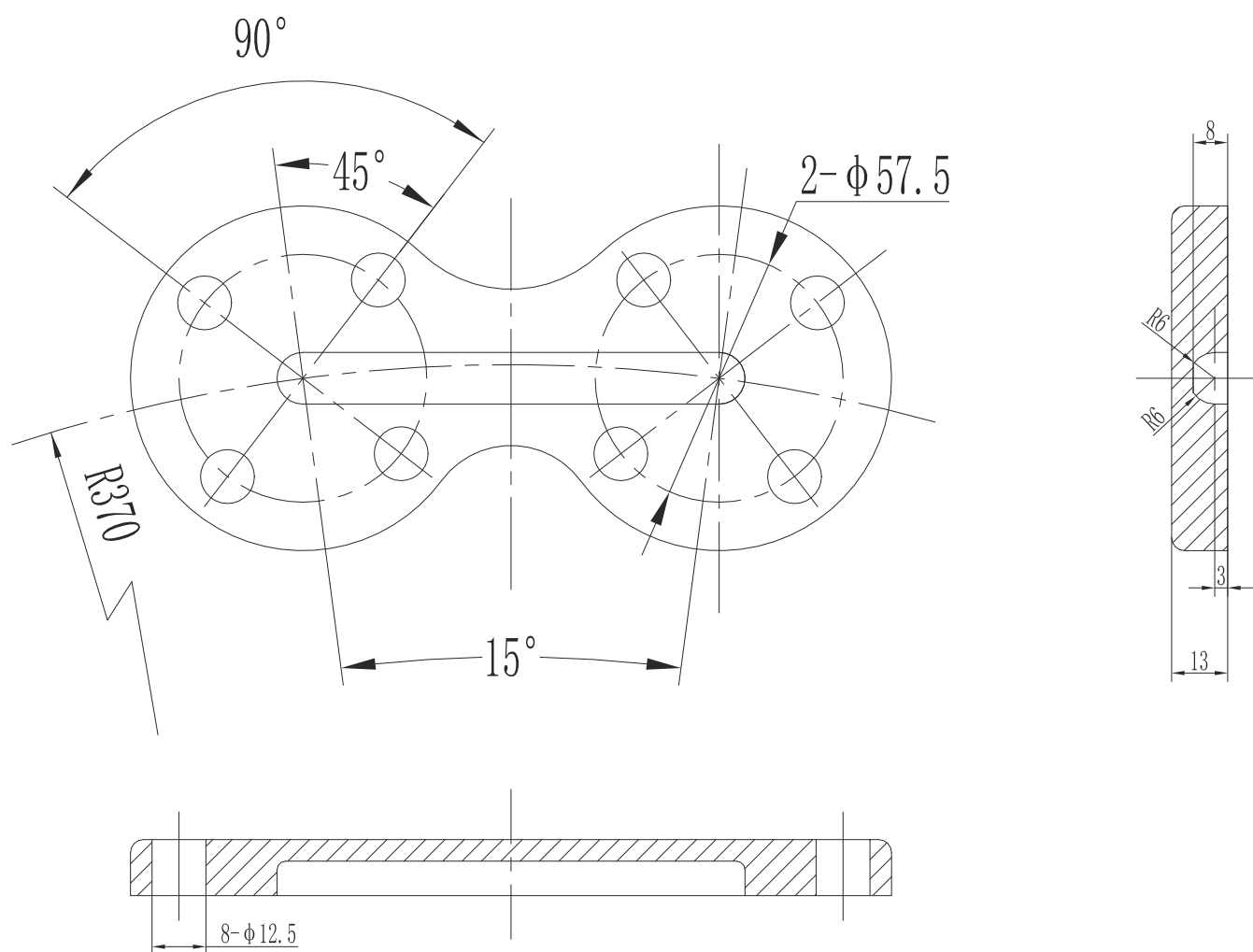


Appendix 4. The Schematic drawing for connection of CV2 and SHM-D



Appendix 5. Schematic drawing for connection of SHZV and CMA-7

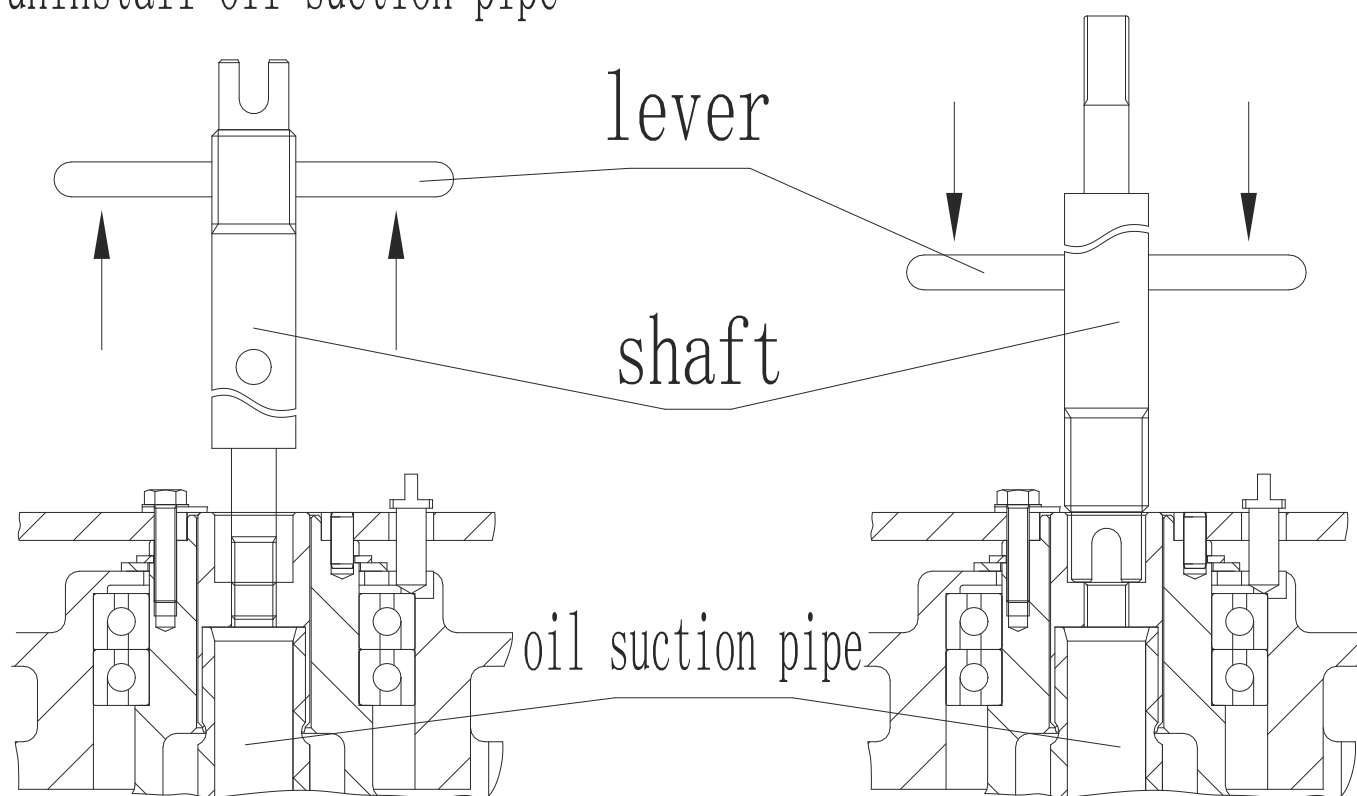


Appendix 6. By-pass pipe, overall dimension

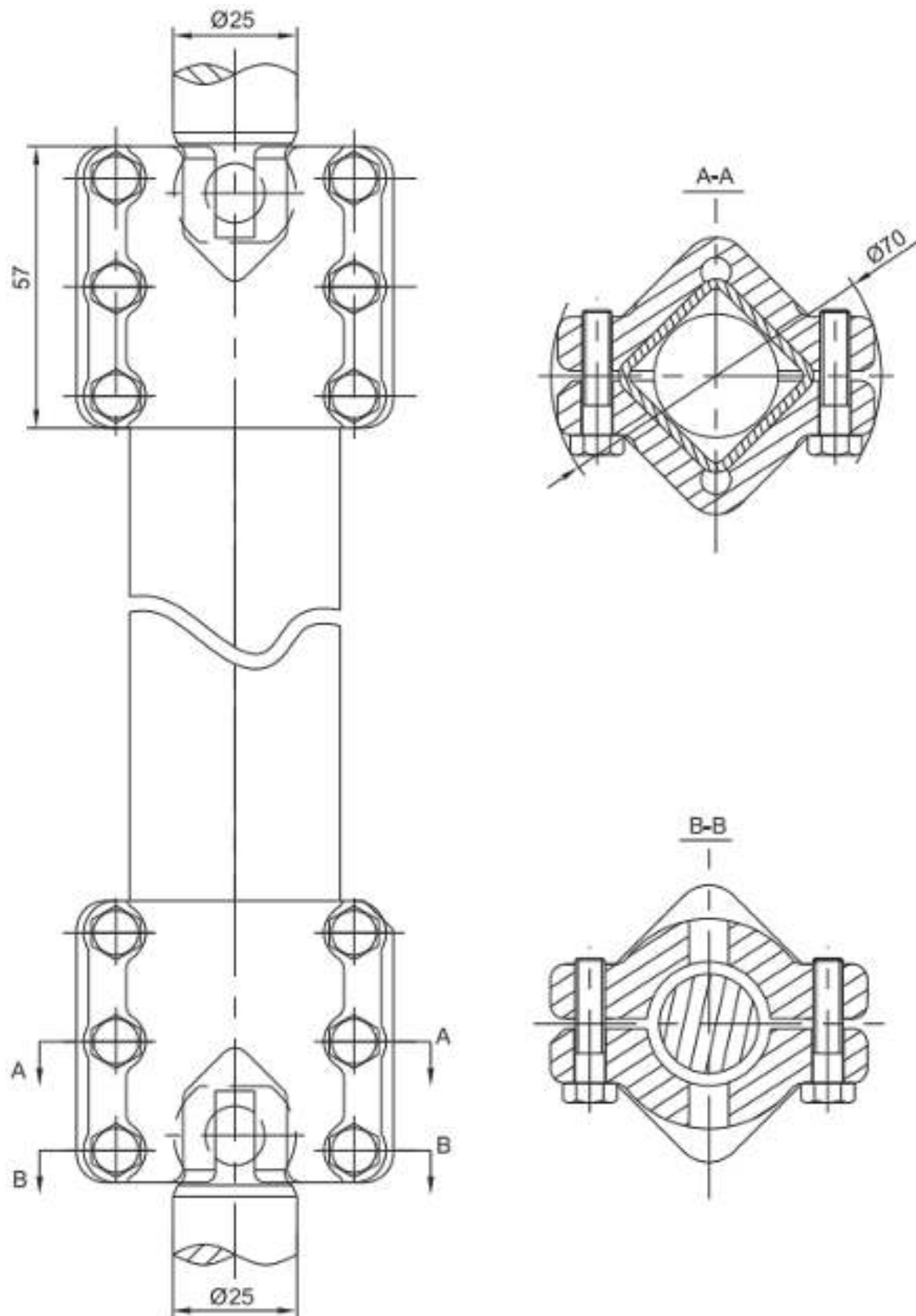
Appendix 7. Special tools for oil drain stoper installation and dismantling

uninstall oil suction pipe

install oil suction pipe

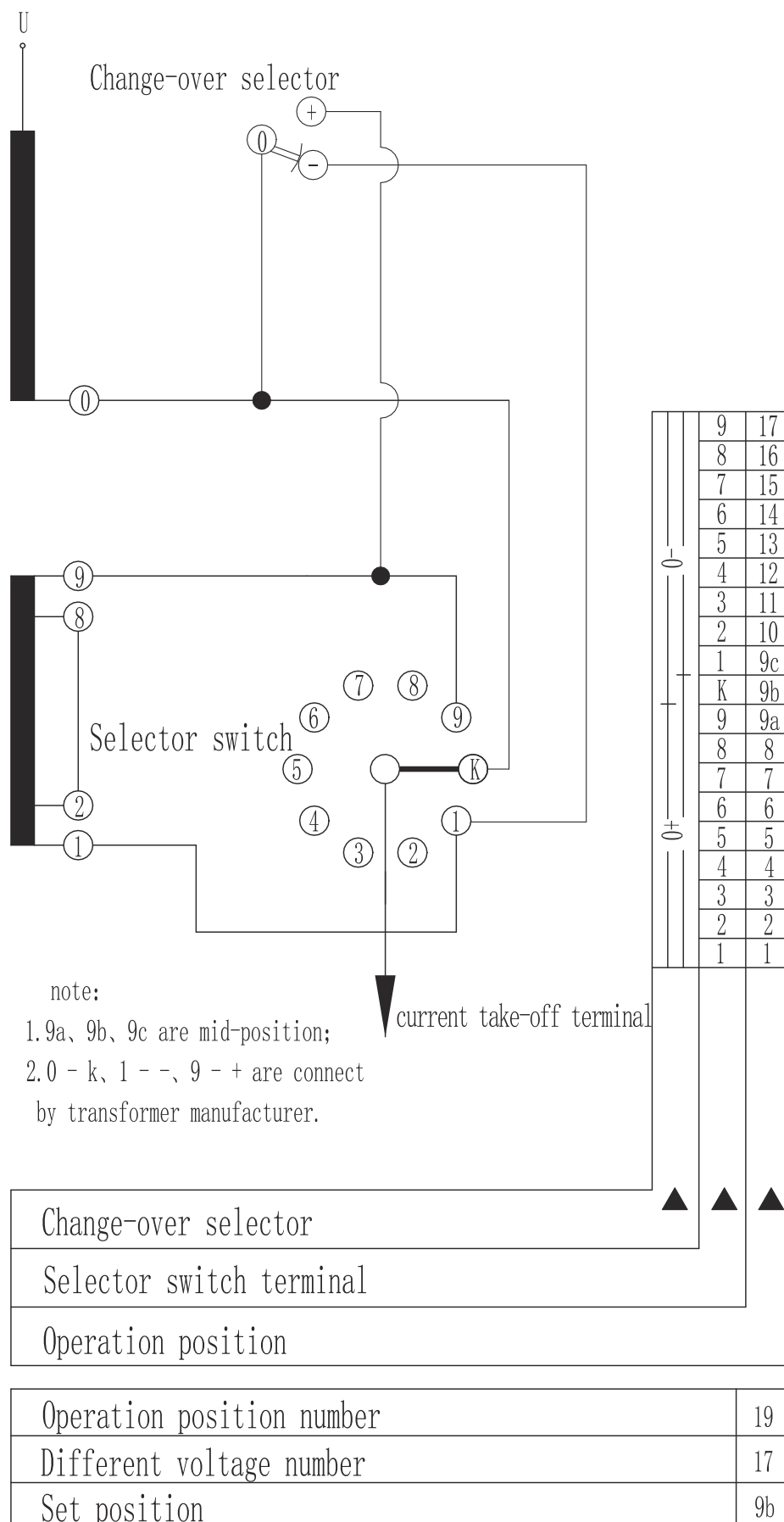


Appendix 8. Schematic drawing for connection of horizontal shaft and vertical shaft

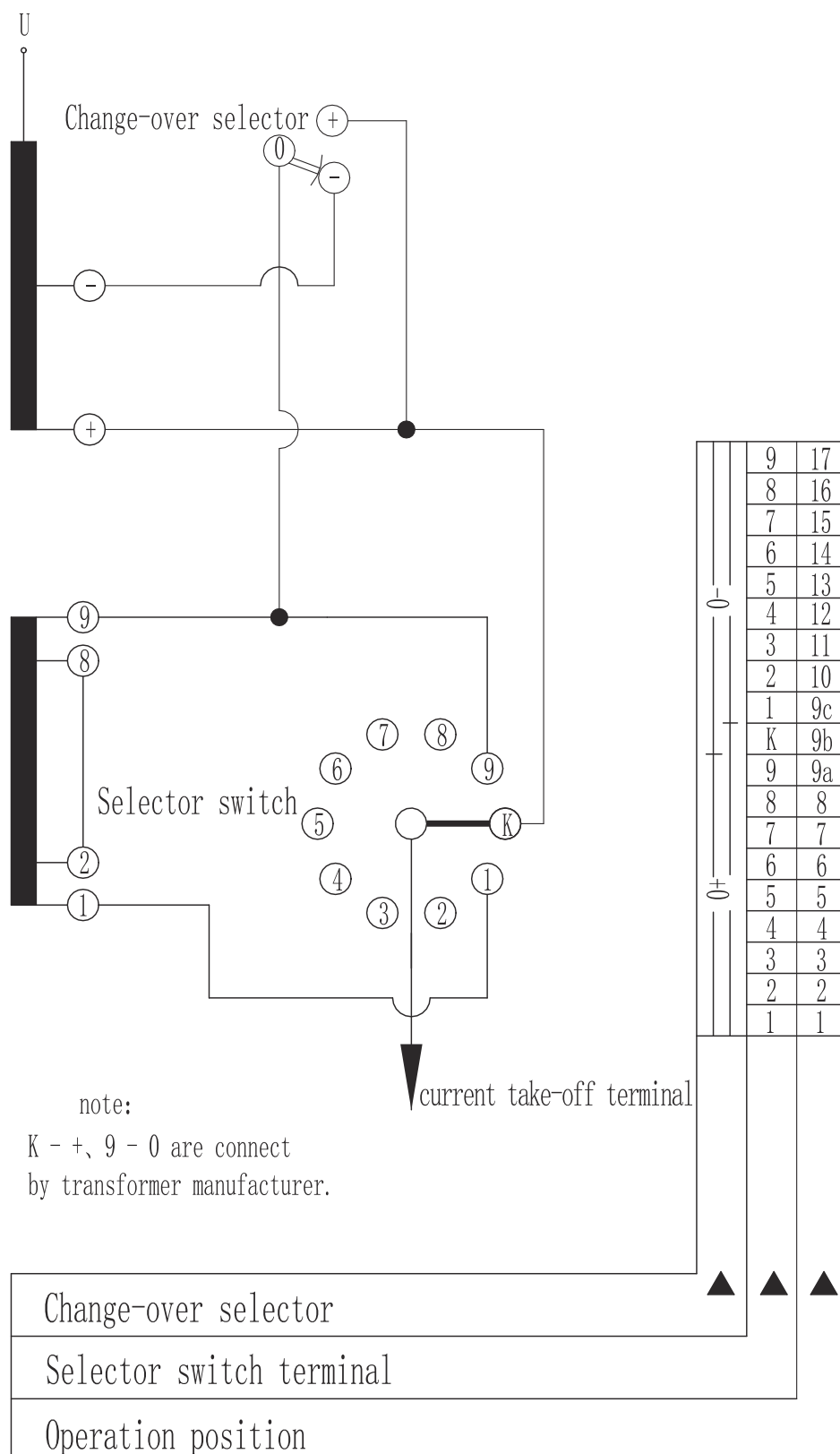


unit: mm

Appendix 9. CV2(10193W) operating position table and connection diagram

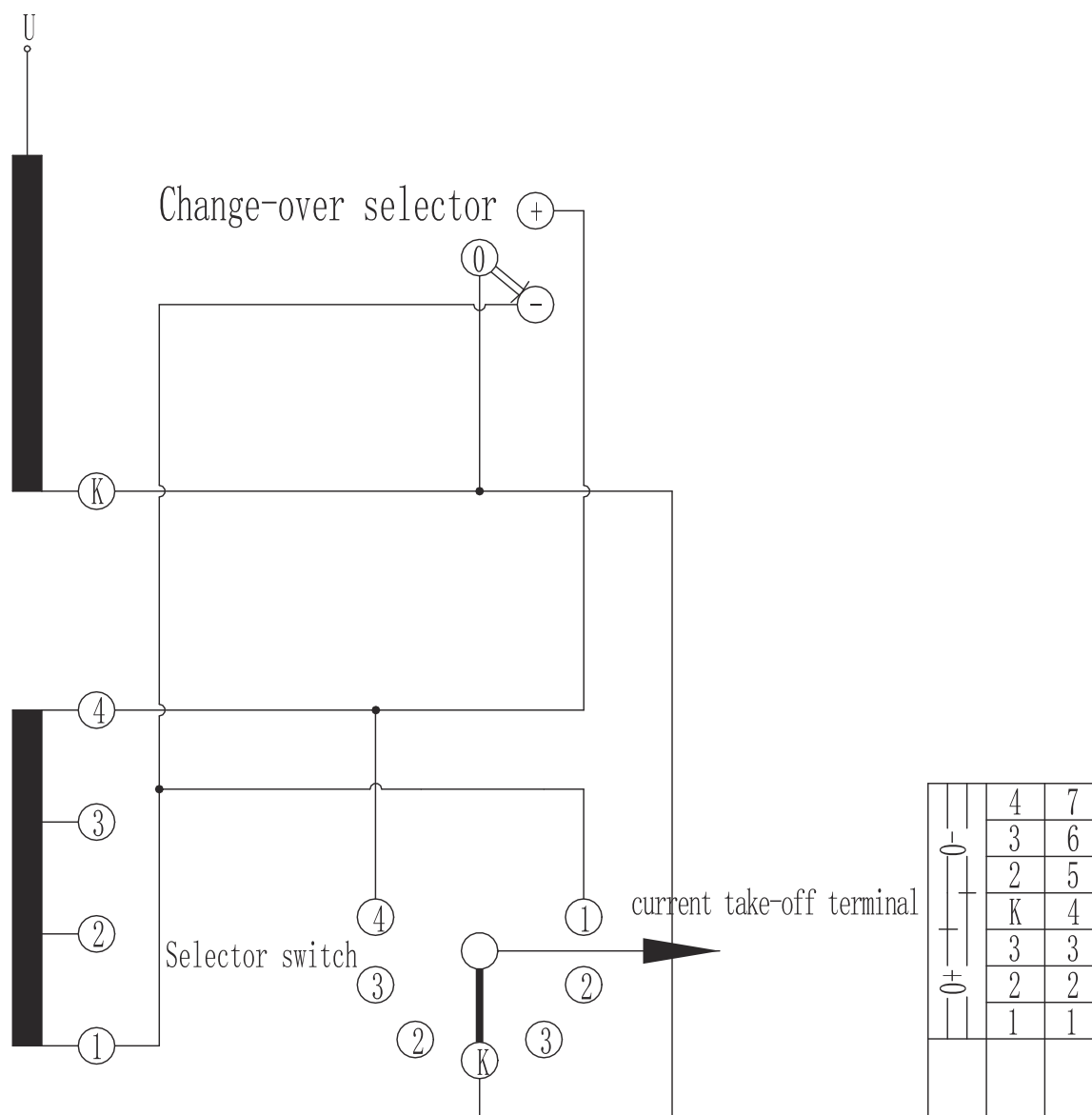


Appendix 10. CV2(10193G) operating position table and connection diagram



Operation position number	19
Different voltage number	17
Set position	9b

Appendix 11. CV2(10071W) operating position table and connection diagram

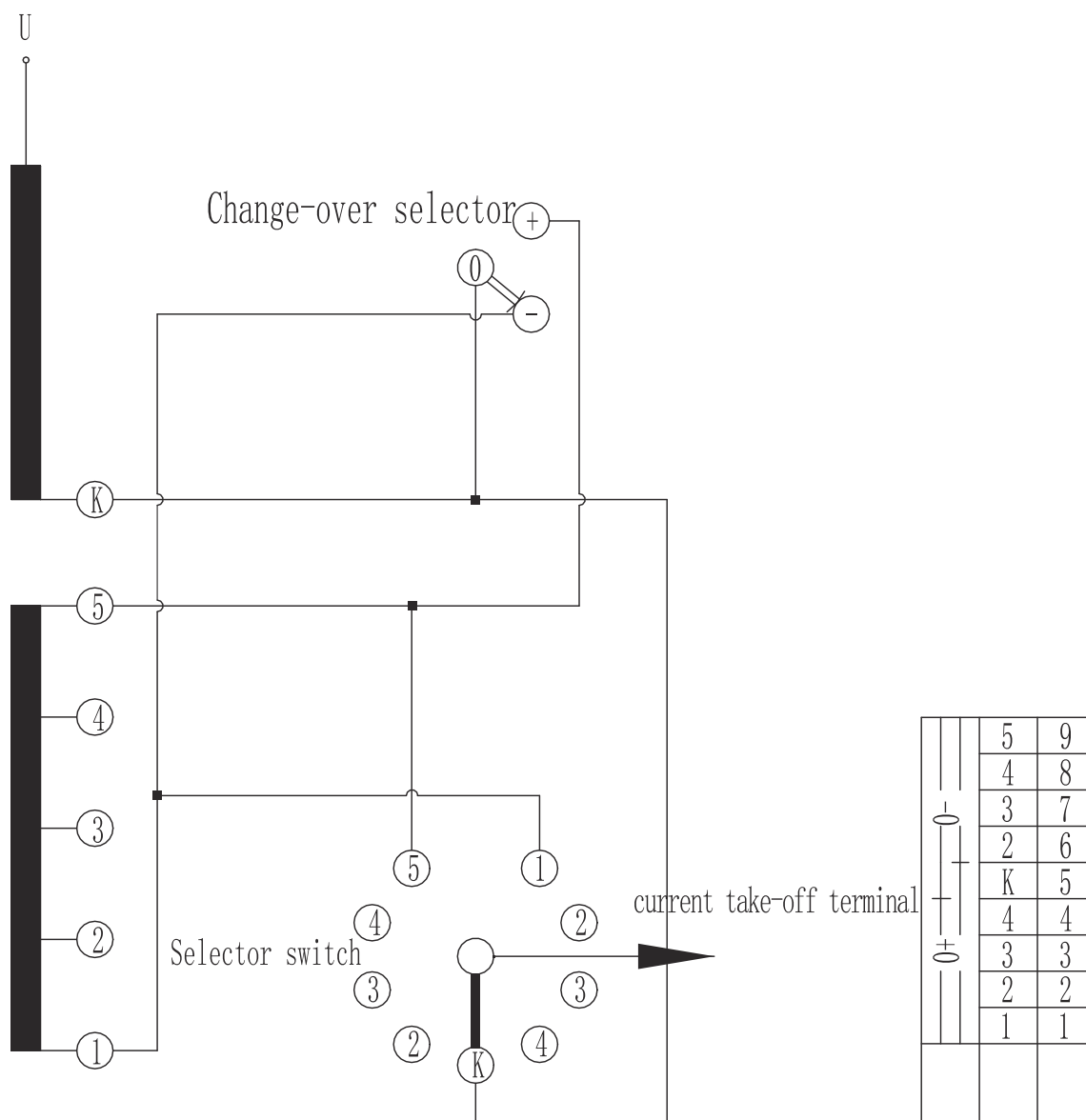


Tap selector contact “+ - 4”、“- - 1”、“2-2”、“3-3”、“0-k” are connect by end user

Change-over selector	▲	▲	▲
Selector switch terminal			
Operation position			

Operation position number	7
Different voltage number	7
Set position	4

Appendix 12. CV2(10091W) operating position table and connection diagram

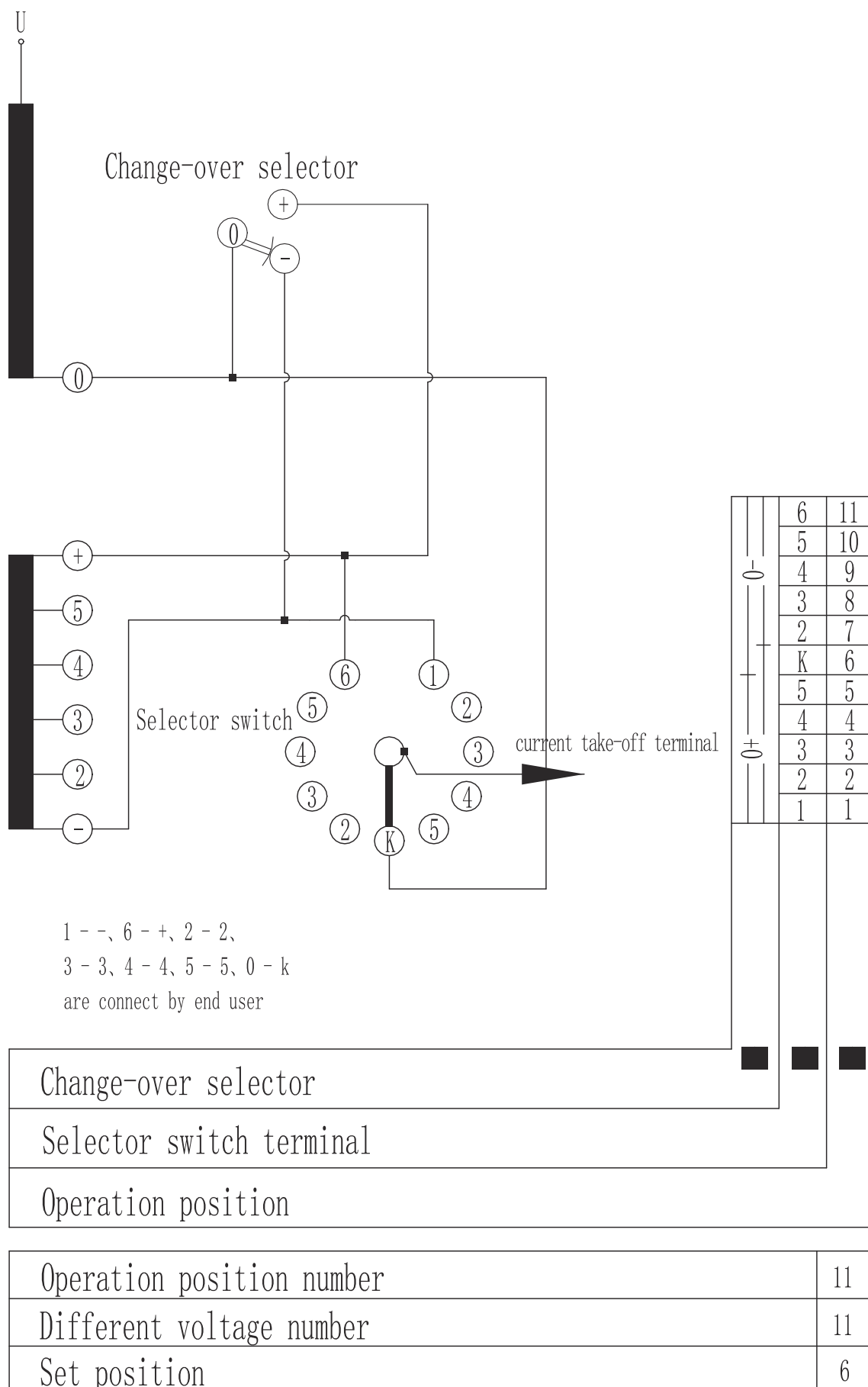


Tap selector contact “+ - 5”、“- - 1”、“2-2”、“3-3”、“4-4”、“0-k” are connect by end user

Change-over selector	
Selector switch terminal	
Operation position	

Operation position number	9
Different voltage number	9
Set position	5

Appendix 13. CV2(12111W) operating position table and connection diagram





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